Scientific African 25 (2024) e02305

Contents lists available at ScienceDirect

Scientific African

journal homepage: www.elsevier.com/locate/sciaf

Optimal selection and design of grid-connected hybrid renewable energy system in three selected communities of Rivers State

Chidozie Ezekwem [a,*], Suresh Muthusamy [b], Peace Chiamaka Ezekwem [c]

[a] Department of Mechanical Engineering, University of Port Harcourt, Port Harcourt, Nigeria
[b] Department of Electrical and Electronics Engineering, Kongu Engineering College, Perundurai, Erode, Tamil Nadu, India
[c] Department of Preventive and Social Medicine, University of Port Harcourt, Port Harcourt, Nigeria

ARTICLE INFO

Editor: DR B Gyampoh

Keywords:
PV modules
Wind turbine
Rural electrification
HOMER
Multi-criteria decision making

ABSTRACT

The pursuit of electrifying rural areas has been a multi-decade quest, emblematic of Nigeria's longstanding efforts to diversify its energy sources and reduce dependence on traditional fossil fuels. Accordingly, this study aims to evaluate the electrification status and perform a techno-economic analysis of a hybrid system to ensure electricity reliability, bill reduction and reduced grid demand for three representative rural communities in Rivers State using HOMER Pro Software. Subsequently, five hybrid system scenarios that encompassed different combinations of the grid, PV modules, wind and battery, were examined using the CRITIC-PROMETHEE II methodologies. Each community was assessed based on its own hourly electricity demand, using data obtained from the injection substation. The Aluu, Choba and Rumuekini communities experienced an average daily electricity supply of 4.6, 5.4 and 6.2 h respectively from the national grid with energy usage of 5436.4 MW for Aluu, 3375.4 MW for Choba and 1626.1 MW for Rumuekini, in 2022. The findings indicate that the grid/PV module was the optimal choice for the three communities. Moreover, the cost of energy (COE) for the proposed systems ranged from 0.0181 \$/kWh to 0.0185 \$/kWh, significantly lower than the grid's COE of 0.08625 \$/kWh in the respective communities. This research further includes a sensitivity analysis of the optimal system, examining variations in PV cost, discount rate, inflation rate, annual scaled solar irradiance and electrical demand. The results provide a comprehensive framework to support decision-making, optimize system design and align renewable energy solutions with project goals and stakeholder priorities.

Introduction

Nigeria, situated on the Gulf of Guinea in sub-Saharan Africa, is home to a populace of approximately 200 million [9,10]. Nigeria should be considered as the frontrunner on the continent. As the most populous nation in Africa, it boasts the largest economy, and is a leading oil producer. Yet, Nigeria is besieged by massive developmental challenges, which have resulted in poverty and conflict in the country. Encumbrances to Nigeria's economic potential include infrastructure deficiencies which demand significant upgrades in critical areas like power supply and roads. Escalating insecurity, coupled with pervasive underemployment and unemployment, further compounds these challenges. Effectively addressing these issues necessitates a comprehensive approach that involves efficient

* Corresponding author.
E-mail address: chidozie.ezekwem@uniport.edu.ng (C. Ezekwem).

https://doi.org/10.1016/j.sciaf.2024.e02305
Received 19 March 2024; Received in revised form 25 June 2024; Accepted 1 July 2024
Available online 5 July 2024

Nomenclature

AC	Alternating Current
AHP	Analytical Hierarchy Process
COE	Cost of energy ($/kWh)
CRF	Capital recovery factor
CRITIC	Criteria importance through intercriteria correlation
DC	Direct Current
HOMER	Hybrid Optimization Model for Electric Renewable
IRR	Internal rate of return
LCOE	Levelized cost of energy ($/kWh)
MCDM	Multiple criteria decision making
MD	Maximum Demand
NASA	National Aeronautics and Space Administration
NERC	Nigerian Electricity Regulatory Commission
Non-MD	Non-Maximum Demand
NPC	Net present cost ($)
O&M	Operation and maintenance ($)
PHEDC	Port Harcourt Electricity Distribution company
PROMETHEE	Preference Ranking Organization Method for Enrichment of Evaluations
PV	Photovoltaic
RF	Renewable fraction
ROI	Return on Investment
Uniport	University of Port Harcourt
UPTH	University of Port Harcourt Teaching Hospital
CRF	Capital recovery factor
B_c	Battery capacity (kWh)
D_A	Daily autonomy
DOD	Depth of discharge
$C_{ann.tot}$	Annual total cost ($/yr)
$C_{i,ref}$	Nominal annual cash flow for base (reference) system,
C_i	Nominal annual cash flow for current system,
C_{cap}	Capital cost of the current system
$C_{cap,ref}$	Capital cost of the base (reference) system.
CF_t	Cash is flow at time t
$E_{prim.AC}$	AC primary load served,
$E_{prim.DC}$	DC primary load served
$E_{grid.sales}$	Total grid sales
$E_{Load,\ d}$	Daily load consumption (kWh/day)
f_{PV}	PV derating factor (%)
f	Annual inflation rate
G_T	Solar radiation incident on the PV array in the current time step (kW/m^2)
G_o	Incident radiation under standard test conditions (1 kW/m^2)
i	Interest rate (%)
i_o	Nominal interest rate
N	Number of years
n	Number of time periods.
P_o	Wind turbine power output at standard temperature and pressure (kW)
P_{PV}	Output power (kW)
P_{Rc}	Rated capacity of the PV panel under standard test conditions (kW)
R_{proj}	Project lifetime in years
T_c	PV cell temperature in the current time step (°C)
T_o	PV cell temperature under standard test conditions (=25 °C)
U_{hub}	Wind speed at the hub height of the wind turbine (m/s)
U_{anem}	Wind speed at the hub height of the anemometer (m/s)
z_{hub}	Hub height of the wind turbine (m)
z_{anem}	Anemometer height (m)
z_o	Surface roughness length (m)
ρ_a	Actual density of air (kg/m^3)

ρ_o	Air density at standard temperature and pressure (kg/m^3)
β	Temperature coefficient of power (%/ °C)
β_{INV}	Inverter efficiency (%)
β_{VRFB}	Battery efficiency (%)
$\bar{r}$	Arithmetic mean
$\varphi^+(a)$	Leaving (positive) outranking flows,
$\varphi^-(a)$	Entering (negative) outranking flows
σ_j	Standard deviation

governance, targeted policies, substantial investments in both infrastructure and human capital, and measures to tackle corruption head-on. Nigeria has a long way to go in achieving electricity sufficiency. At present, its national grid, which currently supplies 51.2 % of the nation's electricity, struggles to match the escalating demand for electricity [47]. According to the report of World Bank, as of 2018, 56.5 % of Nigerians lacked access to grid electricity, with the figure dropping to 55.4 % in 2020 and 59.5 % in 2021 [72]. Electricity companies typically focus on providing services to urban areas, neglecting rural populations [36]. The same report indicates that, in 2018, 31 % of rural areas lacked grid electricity, with this figure dropping to 24.6 % in 2020 and increasing slightly to 26.3 % in 2021 [73]. Consequently, the current energy services in Nigeria fall short of meeting the needs of rural communities. The lack of reliable and high-quality electricity perpetuates poverty, hinders essential social services, and limits opportunities for impoverished populations. Hence, the rural electrification is paramount to both the social and economic development of Nigeria. The national grid failed multiple times, with 222 system collapses recorded between January 2010 and June 2022 [65]. A major factor contributing to these grid failures in Nigeria is the insufficient capacity of electricity generation relative to the growing demand. Furthermore, Nigeria's rapidly expanding population has also put strain on the country's current grid work, underscoring the imperative to integrate renewable energy sources for enhancing system stability and reliability. This transition not only addresses the problems posed by population growth but also aligns with global efforts to combat climate change and promote sustainable development. In this context, renewable energy sources are poised to play a pivotal role in the future of electricity generation. Nigeria is incredibly fortunate to have abundant renewable energy resources, offering significant potential to achieve net-zero emissions. The shift towards renewable energy sources can reduce reliance on dwindling fossil fuels and energy imports, major contributors to greenhouse gas emissions. Therefore, it is imperative to focus on developing and diversifying the energy system mix. The Nigerian government, through the National Renewable Energy Action Plan (NREAP), aims to increase the share of renewable energy in the electricity mix to 15 % by 2030, excluding medium and large hydropower. Presently, the exiguous share of renewable energy is 0.17 % (with solar at 0.12 % and bioenergy at 0.05 %), which is negligible despite the country's vast renewable energy potential, particularly in wind and solar [62]. Achieving this ambitious target necessitates more aggressive strategies for transforming energy production. One significant barrier to the large-scale implementation of renewable energy in Nigeria is the inadequacy of research initiatives [42]. Robust research and development are crucial for advancing renewable energy technologies, optimizing their deployment, and addressing region-specific challenges.

Solar and wind energy stand out as the most promising and sustainable renewable energy resources [21,37]. These prominent renewable sources are widely used for electricity generation globally, playing a crucial role in the transition to a low-carbon and sustainable energy future [22]. Many regions of the globe have relied on wind and solar power as their primary means of producing electricity. For instance, the United States boasts significant solar power capabilities, including extensive solar farms situated in the Mojave Desert, alongside numerous wind farms. The Bhadla Solar Park stands out as one of the largest solar parks globally, boasting an installed capacity exceeding 2245 MW [43] while Gansu Wind Farm, located in China's Gansu province, stands as one of the largest wind farms worldwide, surpassing a total capacity of 6000 MW. Africa is also making huge investments in renewable energy infrastructure. The Benban Solar Park, located near Aswan in Egypt, is the largest solar park in Africa, with a total capacity of roughly 1.8 GW when finished. The Lake Turkana Wind Power Project, located in northern Kenya, is the biggest wind farm in Africa, with a capacity of 310 MW. These projects stand out as monumental renewable energy developments in Africa and reflect significant investments and efforts aimed at harnessing renewable resources, reducing carbon emissions, and increasing energy access on the continent. Nigeria, being geographically close to the equator places it in a very high sun belt, with solar irradiation values ranging from approximately 3.54 to 7.01 kWh/m^2/day. This positioning renders solar power a feasible and attractive renewable energy option for integration into the national grid [28,66]. Its average wind speed varies between 3.0 m/s in the south regions and up to 7.5 m/s in the northern parts of the country due to the dry and more arid climate [13]. Despite these abundant renewable resources, the share of electricity generated from renewables in Nigeria declined significantly, dropping from 23.3 % in 2011 to 16.4 % in 2022 [63]. The country's energy sector employs gas as its major energy source for electricity generation in order to offer the populace with affordable power. However, gas rich Nigeria can't find enough for its power plants making the country's power sector unstable and insecure that have affected mostly rural communities. Nigeria's electricity generation capacity is approximately 12,552 MW, but the actual available capacity is often much lower, around 4000 MW [49]. In contrast, Egypt, another developing country with a population half the size of Nigeria's, generates over 50,000 MW [38]. Recent events, such as the Russian-Ukraine standoff, have indeed highlighted vulnerabilities in global energy markets significantly impacting countries like Nigeria that depend heavily on imported refined petroleum products to satisfy domestic fuel demands. The resulting spike in fuel prices is crippling economies, especially for a country like Nigeria already grappling with recession and the impacts of the COVID-19 pandemic. According to a World Bank report, businesses in Nigeria lose $29 billion annually due to unreliable electricity, which is equivalent to about 2 % GDP [19]. Thus, leveraging alternative energy sources like wind and solar presents a viable solution to mitigate these challenges.

HOMER (Hybrid Optimization Model for Multiple Energy Resources) is a powerful computational software tool that enables the optimization and analysis of grid-dependent and off-grid energy generation systems. It is widely recognized and used by engineers, scholars, and planners to design and evaluate renewable energy-based power systems, in order to determine optimal size of its components [14]. Therefore, research on the techno-economic implications of hybrid renewable energy power generation is essential for accelerating the transition to clean and sustainable energy sources. Such research can inform policy decisions, drive technological innovations, and provide valuable insights into the benefits and challenges of integrating multiple renewable energy sources into the energy mix. In the past, many researchers have examined the technical and economic viability of energy systems for providing electricity to rural areas in different parts of Nigeria using HOMER software. Adaramola [2] determined the optimal size of a PV power system connected to the grid for the case of Jos, Plateau State, with an LCOE of $0.103/kWh. In a study conducted by Salisu et al. [59], the feasibility of a hybrid PV/Wind/Diesel/Battery system for supplying electricity to Giri village in Gwagwalada, Nigeria, was thoroughly examined. The study revealed the PV/Diesel/Battery system as the most economically suitable configuration with COE of $0.110/kWh. Okonkwo et al. [50] conducted a study on the electrification needs of a rural community (Umuejim, Amorka) in Nigeria. The study concluded that a renewable energy system utilizing PV and battery technology was the most economically feasible option. More recently, Odetoye et al. [46] proposed a hybrid standalone renewable microgrid designed to supply electricity to Arandun, Kwara State. This hybrid system combines solar PV, CSP, micro-hydro and wind energy with a battery storage system to achieve an optimal balance capable of meeting the load on a multiyear basis without being assisted by an external grid. The uniqueness of their work lies in the innovative use of HOMER Software, particularly through the integration of a CSP module into the energy mix. Table 1 presents prior research on energy systems utilizing HOMER, with Nigeria as the central focus. While the majority of prior research in Nigeria utilizing HOMER has focused on off-grid systems to provide electricity access in the remote villages, there is a burgeoning necessity to delve into grid-connected applications, as indicated by the limited number of studies in Table 1. Rajbongshi et al. [54] highlighted that the COE for a grid-connected hybrid system is higher than that of an off-grid hybrid system with similar load patterns. Accordingly, they investigated the design of hybrid systems incorporating PV, biomass gasifier, diesel and grid connection to electrify Jhawani village in India. This design successfully reduced the COE from $0.145/kWh to $0.064/kWh for both on-grid and off-grid scenarios. The integration of renewable energy into power grids is foundational to modern energy systems, offering a pathway toward achieving sustainability, resilience, and economic viability in the global energy landscape.

The primary challenge in utilizing HOMER lies in generating a realistic load profile to provide practical optimal designs, especially for community-scale applications. By incorporating actual load profiles, HOMER can more precisely size power generation systems to meet the specific energy needs of users. This approach helps prevent over- or under-sizing of components, thereby enhancing system performance while reducing costs. In Nigeria, certain studies have relied on computed or estimated load profiles, which may not accurately represent true energy consumption patterns. Sofimieari et al. [61] conducted a techno-economic evaluation of renewable electricity for a remote community in Kaduna State, northern Nigeria. Their approach to load analysis involved using typical daily load profiles, which consisted of 24-hour energy demand time series repeated 365 times to simulate an entire year. Furthermore, their study focused on a remote community comprising 200 buildings, where they estimated the energy load based on one representative building and extrapolated it to the entire community. However, the diverse characteristics among buildings within a community—including variations in size, occupancy patterns, energy usage behaviours, and the presence of energy-intensive appliances—highlight the potential limitations of using a single building's load as a basis for the entire community. Adebanji et al. [4] conducted a study to assess

Table 1
Previous studies of hybrid energy systems by using HOMER using Nigeria as the point of location.

Author	System type	Hybrid Options					Application (size)	Evaluation area
		PV	Wind	DG	Battery	Biogas		
[2]	On-Grid	√					Community (300 households)	Jos, Plateau State
[3]	Off-grid	√		√	√		Community (1500 households)	Jos, Plateau State
[52]	Off-grid	√	√	√	√		Community (40 households, a primary health centre, a public primary school, 3 shops, a community hall and few miscellaneous loads were assumed in each location)	Ngo (River state), Ete (Enugu state), Onilu (Lagos state), Zalau (Plateau state), Gubio (Borno state), Tambo (Adamawa state)
[15]	Off-grid	√		√	√		Community (148 houses, a school and a mosque)	Gajiram village, Borno
[41]	Off-grid	√	√		√		Rural communities (200 homes)	Sokoto State
[8]	Off-grid	√	√	√	√		Senate building	Kwara State
[59]	Off-grid	√	√	√	√		Community (80 households, a primary school and a health centre)	Giri village, Abuja
[26]	Off-grid	√		√	√		Community (300 households)	Lade II, Kwara State
[48]	Off-grid	√	√	√	√		University community	Plateau, Kwara, FCT, Kogi, Benue, Niger
[51]	On-grid, Off-grid	√	√	√	√		Community (20 households)	A typical remote village in North-western region of Nigeria
[12]	Off-grid	√	√	√	√		Rural community (40 households)	Abadam, Borno State
[34]	On-Grid	√	√			√	Community (200 households)	Zaria, Kaduna State
[7]	On-Grid	√					Community (24 households)	Ikose, Oyo State

the feasibility and optimal design of a hybrid power system for rural electrification in Itapaji-Ekiti, Nigeria. Their research utilized electricity consumption data gathered through questionnaires. However, this data collection approach introduces potential inaccuracies or biases in responses from participants, which could distort the data. Furthermore, questionnaires may not fully capture the complexity of energy usage patterns, resulting in incomplete or unreliable data. Mohammed et al. [44] conducted a feasibility study on a standalone Wind/PV hybrid energy system intended for off-grid rural electrification in Sokoto State, Nigeria. However, their study lacked detailed load estimation for the 50 houses, failing to provide specific values for the variability across different days and months. These methods of load estimation often overlook the dynamic interplay between renewable energy availability and energy demand, as well as the inherent variability in load, particularly during peak periods or sudden shifts in consumption rates. Such oversights could lead to suboptimal system designs or resource allocations [35].

Multiple Criteria Decision Making (MCDM) plays a crucial role in determining the most appropriate hybrid renewable energy systems. In hybrid renewable energy systems selection process, various assessment criteria and alternatives must be evaluated concurrently, taking into account their interconnections and potential trade-offs. MCDM methods enable the evaluation of these and other conflicting factors, as well as the determination of the most suitable alternative based on various criteria. In existing literature, several studies have explored the combination of MCDM with HOMER methodologies to evaluate renewable energy technologies. Jahangiri et al. [31] utilized the fuzzy MCDM technique to optimize a hybrid micro-grid system across five stations in Qatar, assessing the potential for generating electricity and hydrogen through renewable solar-wind energy sources. Their findings indicated that Doha International Airport is the most suitable site for constructing a PV/Wind hybrid energy generation system. Similarly, Ali et al. [6] proposed a new hybrid framework using an integrated CRITIC—CODAS model to determine the optimal hybrid energy system for coastal off-grid areas in Bangladesh. They concluded that a PV/Battery/Biogas/DG system was the best solution for the study area, with an investment cost of $75,688 and a COE of $0.28/kWh. He et al. [29] employed HOMER software to determine the optimal design for hybrid renewable systems, including PV, bio generator, diesel generator, and battery configurations for both on-grid and off-grid scenarios. The optimal scenario, selected using TOPSIS and considering economic, environmental, and technical factors, was a PV/Bio-Diesel/Battery system with an NPC of $1.02 million and a COE of $0.188/kWh. Additionally, Ullah et al. [69] applied an integrated hybrid F-AHP/MOORA/TOPSIS approach to select the best renewable energy supply system for Ghulam Shah Township in Pakistan, considering both on-grid and off-grid modes. John et al. [33] employed the Analytical Hierarchy Process (AHP) to evaluate five renewable energy sources (PV, solar thermal, wind, biomass, and geothermal) in Saudi Arabia. Similarly, Salameh et al. [58] introduced a hybrid MCDM framework in Neom City, Saudi Arabia, to determine the optimal hybrid renewable energy system for industrial applications. Their approach began with a feasibility analysis of various alternatives using HOMER software, followed by weighting 16 criteria with methods such as no priority, CRITIC, and Entropy. The ranking of nine different alternatives was conducted using TOPSIS, WASPA, MOORA, and EDAS. The findings indicated that the best option comprised solar PV, diesel generators, and battery energy storage. Alanazi et al. [5] integrated AHP and Simple Additive Weighting (SAW) methods to evaluate PV and solar thermal power technologies across three different cities in Saudi Arabia. In another study, Rezk et al. [55] utilized the CRITIC and TOPSIS procedures to ascertain the optimal energy management strategy for a hybrid PV/Diesel battery-based water desalination system in isolated regions, considering technical, economic, and environmental criteria. Ukoba et al. [68] conducted an optimization study on hybrid renewable energy systems to meet the energy needs of an average household across Nigeria's six geopolitical zones. They began with a feasibility analysis using HOMER software, followed by criteria weighting using the AHP and ranking alternatives with the TOPSIS. Their results revealed that the Biomass generator/solar PV/Battery energy system was the most effective across all zones. Similarly, Diemuodeke et al. [24] carried out a feasibility analysis to determine the optimal configuration of a PV/Wind/-Diesel/Battery hybrid renewable energy system for household electrification in six locations within Nigeria's south-south geopolitical zone. They ranked various hybrid renewable energy system configurations using the TOPSIS method, taking into account technical, economic, environmental, and sociocultural factors. They found that the COE for reliable and sustainable configurations ranged from $0.459/kWh to $0.562/kWh. However, from literatures, the use of multiple criteria in system ranking have received low research focus in the context of rural community electrification, which highlights a significant gap in the literature. PROMETHEE (Preference Ranking Organization Method for Enrichment of Evaluations) is a crucial decision-making tool used when dealing with multiple conflicting criteria and alternative actions. PROMETHEE II builds upon its predecessor, PROMETHEE I, offering several advancements. Compared to PROMETHEE I, PROMETHEE II provides a stronger mathematical foundation and incorporates preference functions to determine preference degrees between alternatives based on the differences in evaluations for each criterion. This enhances the precision and accuracy in expressing preferences. PROMETHEE II allows for a more refined measurement of preferences by considering both the strength and direction of preferences between alternatives for each criterion, offering a detailed insight into decision-makers' preferences. Additionally, PROMETHEE II establishes partial preorders for alternatives based on preference degrees, enabling a comprehensive analysis of relative rankings on individual criteria. According to Ridha et al. [57], PROMETHEE II demonstrates high stability in decision-making scenarios. However, the use of PROMETHEE II methodologies for analyzing renewable energy remains limited, with Chisale et al. [18] evaluating the optimal configuration of grid/PV/Biogas HRES for design and simulation in a school setting using the integrated CRITIC-PROMETHEE II approaches.

The reviewed literature establishes a strong basis for this study by exploring technical and economic case studies from diverse global locations. However, a comprehensive viability study of grid-integrated solutions for enhancing rural electrification, particularly in Rivers State, Nigeria, remains inadequately addressed. Therefore, this study seeks to identify the optimal combination of renewable energy technologies, leveraging available resources to reliably and sustainably meet the electricity demand in specific communities within Rivers State. This project is innovative in its design of grid-connected hybrid systems utilizing renewable energy sources, aiming to improve access to high-quality, reliable electricity in these rural areas. Grid-connected generation plays a crucial role in addressing energy shortages by increasing the overall capacity of the electricity grid and optimizing power resource distribution [54].

Additionally, this study is a pioneer in applying the CRITIC and PROMETHEE II methodologies to identify the optimal hybrid energy system within the Nigerian context. Another novel aspect of this research is the load profile, which is derived from real-time data obtained from the power grid. Using up-to-date real-time data from the electricity grid will assist energy policymakers in Nigeria in making more informed decisions. Additionally, the study analyzes the current electricity situation in these regions and suggests that the country should enhance its energy resource development by diversifying its energy portfolio. This approach would not only foster sustainable livelihoods in rural communities but also ensure national energy security and support the efficient operation of the energy delivery system.

Power transmission network in Port Harcourt

Port Harcourt region currently controls six transmission substations: Ahaoda, Port Harcourt Mains, Port Harcourt Town, Yenegoa, Rumuosi, and Elelenwo [27]. Rumuosi transmission station receives its energy from Port Harcourt Mains sub transmission station and Omoku gas turbine at 132 kV. The 132 kV transmission line is stepped down to 33 kV at the Rumuosi transmission station and then transmitted to the following substations in Port Harcourt: UPTH, NTA, Rukpokwu, and New Airport Road. Rumuosi transmission station is equipped with two transformers rated at 132/33 kV 40 MVA (T1) and 132/33 kV 60 MVA (T2), as well as four load feeders: UPTH, NTA, Rukpokwu, and New Airport Road, each of which has a maximum load capacity of 15.6 MW, 13.4 MW, 10.8 MW, and 0.1 MW (Fig. 1). The UPTH lines serve the UPTH, Uniport and Choba injection substations. Subsequently, the Choba injection substations distribute power to the Aluu, Choba, and Rumuekini feeders, which in turn provide electricity to various areas within these communities, forming the focal point of our investigation.

Electricity tariff

The Port Harcourt Electricity Distribution company (PHEDC) is in charge of distributing electricity to a population of 14 million people across four states, namely Akwa Ibom, Bayelsa, Cross River, and Rivers state. In Nigeria, the cost of electricity is fixed and regulated by the Nigerian Electricity Regulatory Commission (NERC). NERC classifies electricity consumers into three main groups: Non-MD, MD-1, and MD-2. Within these categories, there are five distinct bands labelled A, B, C, D, and E, each reflecting varying levels of service and minimum hours of power supply. The electricity rates levied by the PHEDC are detailed in Table 2. As indicated in the table, the current electricity tariff (i.e. with effect from Jan 2023) ranges from N39.44/kWh for E-Non MD to N72.67/kWh for A-Non MD.

Methodology

Study area

Rivers State, Nigeria's 26th biggest state by land, has an area of approximately 11,077 km^2. This research evaluated three Rivers

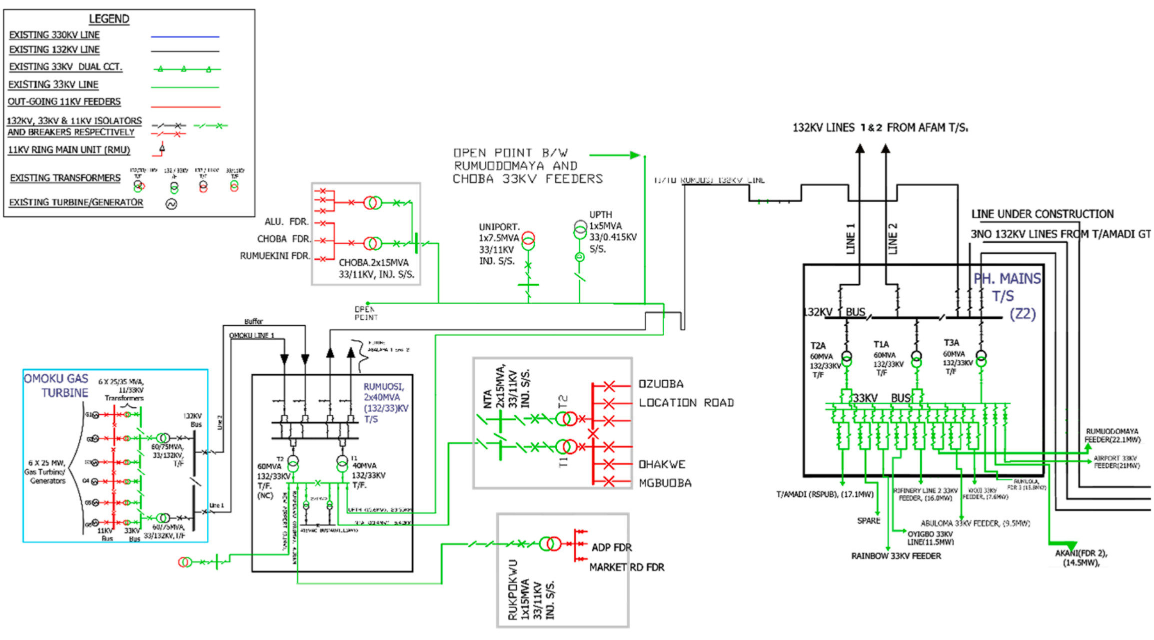

Fig. 1. Diagram of Rumuosi transmission station and injection substations.

Table 2
Applicable tariff for PHEDC.

Tariff bands	Hours of supply	Tariff rate (Naira/kWh)		
		Jan 2022	Feb-Dec 2022	Jan 2023- Till Date
A-Non MD	20–24	57.16	60.67	72.67
A-MD1		56.79	59.64	71.62
A-MD2		54.18	57.94	69.88
B-Non MD	16–20	56.79	59.64	68.96
B-MD1		54.18	57.89	67.18
B-MD2		53.69	57.84	67.12
C-Non MD	12–16	50.15	55.15	56.38
C-MD1		48.45	53.45	54.64
C-MD2		48.45	53.45	54.64
D-Non MD	8–12	35.31	38.81	39.67
D-MD1		50.72	54.22	55.43
D-MD2		50.72	54.22	55.43
E-Non MD	4–8	35.08	38.58	39.44
E-MD1		50.72	54.22	55.43
E-MD2		50.72	54.22	55.43

State communities, Aluu, Choba and Rumuekini, which are indigenous settlements on the fringes of Port Harcourt metropolis, specifically in the north-western part of the city. Aluu is situated to the east of Rumu-Ahunwa and to the southeast of Omuigwe at a coordinate of 4° 56′ 1″ N, 6° 56′ 58″ E, encompassing a land area of 12.4 km^2. The terrain surrounding Aluu is characterized by its luxuriant and verdant nature, owing to its geographical location in the Niger Delta region renowned for its abundant vegetation and water bodies. The terrain around Choba (latitude: 4° 52′ 11″N and longitude: 6° 54′ 31″E) is similar to that of Aluu, with the surrounding area being characterized by tropical rainforest and water bodies. Rumuekini is situated in the geographical coordinates of 4° 53′ 20″ N, 6° 56′ 52″ E. The geographical location of the chosen sites are presented in Fig. 2. These settlements are rapidly expanding due to the presence of the University of Port Harcourt and the abundance of land resources available for housing development. Although, there is access to grid power under the jurisdiction of PHEDC in these communities, but the availability of the power is epileptic and worsens during the raining season. Given the challenges in implementing renewable energy systems, it becomes crucial to select the most appropriate options to ensure optimal resource utilization. This not only enhances the operational efficiency of energy systems but also enables long-term cost savings while improving energy accessibility and affordability.

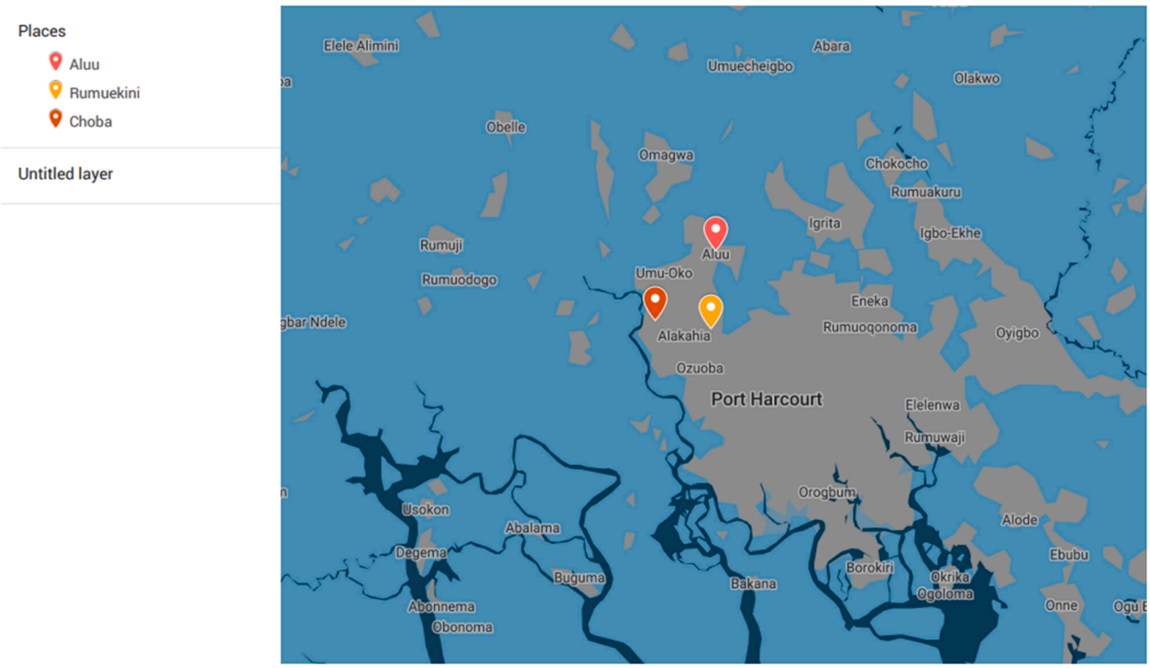

Fig. 2. The geographical location of the selected communities.

System modelling

This study evaluated certain communities located in the south-south region of Nigeria. Electricity supply systems are simulated using the HOMER Pro software. Consequently, the meteorological data (solar radiation, ambient temperature and wind speed) for this investigation were furnished by National Aeronautics and Space Administration (NASA) via HOMER Pro by specifying the latitude and longitude of the chosen locations.

Load assessment

The electrical power requirement of the selected communities were assessed based on a realistic electricity consumption. The load profile analysis was conducted using data gathered from the injection substation covering the case study over the entirety of 2022. Each community is simulated with its own hourly electricity demand. Fig. 3 illustrates the load distribution for each location on a monthly basis. In the case of Aluu, based on this data, the highest average electrical energy consumption occurs in May (5.308 MW), while the lowest occurs in April (1.616 MW). In Choba, the consumption of electrical energy reaches its peak in April, with a value of 2.607 MW, and reaches its lowest point in August, with a value of 1.227 MW. Whereas the use of electrical energy in Rumuekini is highest in March (1.266 MW), and lowest in September (0.509 MW). The hourly energy demand profile of the selected communities can be seen in Fig. 4. Observation of the load profile reveals that power usage is minimal between 1 a.m. and 4 a.m. Following that, the amount of electricity consumed rises from 5 a.m. to 7 a.m. before falling at 8 a.m. The majority of residents in these places spend the most of their time working outside of their homes. Peak demand, on the other hand, occurs in the evening, when the entire family is at home. Based on this, the total annual average energy consumption and peak load for Aluu is 3496.2 kWh/day and 8527.9 kW, with a load factor of 0.41. In Choba, the corresponding figures are 1928.6 kWh/day and 5790.1 kW, with a load factor of 0.33 while Rumuekini is 816.99 kWh/day and 2181.8 kW, with a load factor of 0.37.

Energy resource assessment

The hybrid renewable energy system for electrification was strategically designed according to the available resource potential to meet the community's energy demand. A comprehensive understanding of the renewable energy resources in these communities is essential for the optimal design and sizing of such systems. Solar radiation with the accompanied temperature and wind speed are the fundamental climate data considered.

Available solar radiation and temperature. Fig. 5a presents the changes in the monthly solar irradiation data and mean ambient temperature for the selected communities, sourced from the NASA database via HOMER software, utilizing the coordinates of each location. Due to their close geographical proximity, Aluu, Choba, and Rumuekini typically exhibit similar solar radiation patterns as neighbouring communities. The monthly solar radiation data for these specific locations typically falls within the range of 3.11–5.24 kWh/m^2/day, with an annual average of 4.13 kWh/m^2/day and the average clearness index was found to be 0.416. It is obvious that

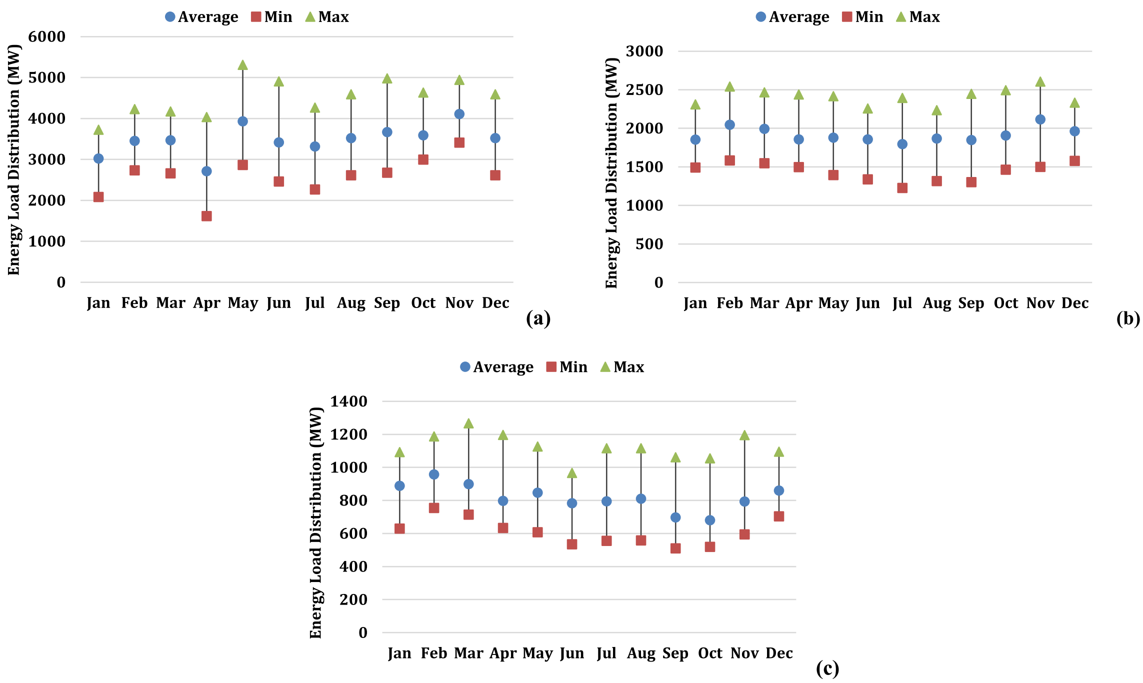

Fig. 3. The three communities, (a) Aluu (b) Choba and (c) Rumuekini and their minimum, maximum, and average monthly load profile.

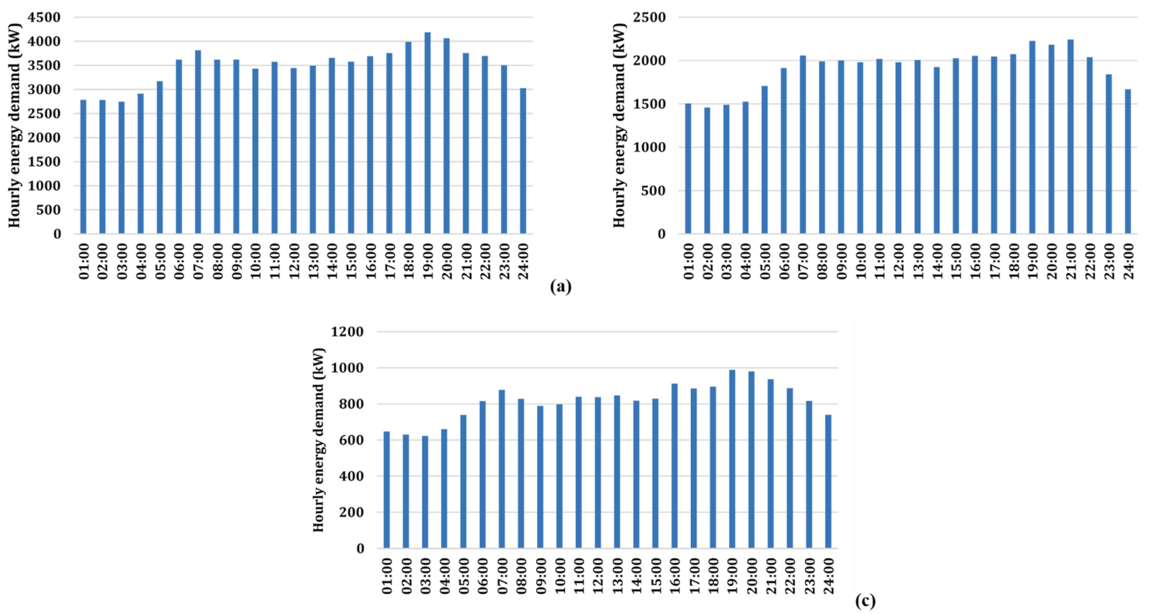

Fig. 4. The three communities, (**a**) Aluu (**b**) Choba and (**c**) Rumuekini and their hourly energy load profile.

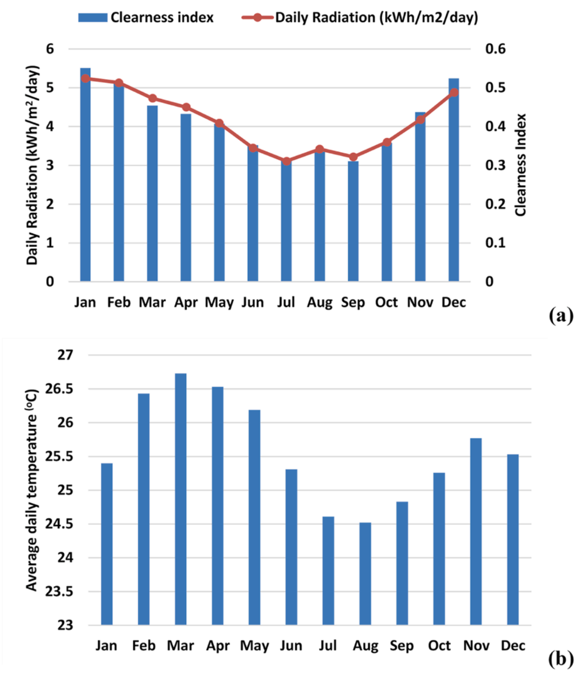

Fig. 5. Average monthly solar irradiation and ambient temperature for the selected communities.

the highest solar irradiance is found between November and February and the lowest rate is found in July. During November to February, Nigeria enters its dry season, marked by clear skies and minimal cloud cover, facilitating optimal sunlight exposure. Conversely, the rainy season, predominant in June and July across many regions in Nigeria, coincides with reduced solar irradiance due to heightened cloud cover and precipitation, limiting the amount of sunlight reaching the Earth's surface and thereby decreasing irradiance levels. Fig. 5b depicts the monthly average ambient temperature for the selected communities. The data reveals that the highest and lowest monthly average temperatures occur in March, reaching 26.73 °C, and in August, dropping to 24.52 °C, respectively. The annual average temperature is found to be 25.59 °C.

Available wind speed. The monthly average wind speed displayed in Fig. 6 was retrieved from the NASA database at 10 m above sea level using an anemometer. The annual averaged wind speed is 3.15 m/s, ranging between 2.43 m/s and 4.00 m/s.

Proposed system architecture

The study examines a system architecture comprising electrical load, converter, battery storages, wind turbine and PV. These components are interconnected with the grid, as depicted in Fig. 7. In the scenario where the system is connected to the grid, the utility grid provides power to fulfil the load demands when the renewable energy sources are unable to meet them, while also efficiently utilizing surplus electricity.

System detail and cost specification

Photovoltaic modules

The photovoltaic module used in this study is a SunPower X21–335W-BLK solar panel, part of SunPower's X-Series. This model features 96 monocrystalline Maxeon Gen III cells, delivering a maximum power output of 335 W. It operates within a temperature range of −40 °C to +85 °C and has a temperature coefficient of −0.29 %/ °C. This panel boasts an excellent module efficiency of 21.1 % and is renowned for its high durability. It has dimensions of 1559 mm x 1046 mm x 40 mm, a power tolerance of ±5 %, a rated voltage of 57.3 V, and a rated current of 5.85A. The capital cost and replacement cost rated at 1 kW, are estimated at $932 with no maintenance cost (SunPower X21–335W-BLK 335 Watt Solar Panel Module (renugen.co.uk)). The PV module is connected to a DC output with a lifetime of 25 years. The de-rating factor considered is 88 % for each panel to accounts for various losses and inefficiencies that occur in a solar power system.

Wind turbine

This study proposed an Enercon E-82 E2, featuring a 3-bladed upwind design and a capacity of 2 MW. Engineered for superior reliability and minimal maintenance, it is well-suited for locations experiencing moderate wind speeds. The configuration employed in this study is grid-connected. With a rotor diameter of 82 m and a hub height of 24.3 m, it is optimized for efficient energy generation. For this project, wind turbines were specified with installed capital cost of $2 million. The replacement cost of the wind turbine considered in this case is $2 million after 20-year service life with an operation and maintenance cost of $30,000 [27].

Converter

In hybrid renewable energy systems, converters play a crucial role in managing and optimizing the flow of electricity between different energy sources, storage devices, and the electrical grid. The primary aim of using converters in these systems is to enable seamless integration, control, and efficient utilization of various renewable energy sources alongside other components within the system. Converters help in transforming and regulating electrical currents to match the requirements of the system and ensure compatibility between different components. The capital cost, replacement cost and O&M costs of the converter for 1 kW system were considered as $550, $550, and $10/year respectively [30]. The operational lifetime of the converter is 15 years, inverter efficiency of 95 % and rectifier efficiency of 55 %.

Battery storage

The primary aim of using battery storage devices in these systems is to address the intermittent nature of renewable energy sources (solar and wind power) by storing excess energy generated during peak production periods and supplying it when there's a demand but insufficient generation. The selected battery energy storage system utilizes lithium-ion technology, which provides a reliable, efficient, and scalable solution for energy storage in electricity generation. With their high energy density, long cycle life, fast charging capabilities, and compatibility with renewable energy sources, lithium-ion batteries are essential for the transition to a more sustainable and resilient energy system. In the context of this research, a 1 kWh lithium-ion battery has been selected. The battery system has a capital cost of $139, a replacement cost of $120 as the prices of batteries are expected to decline further in coming years, and an

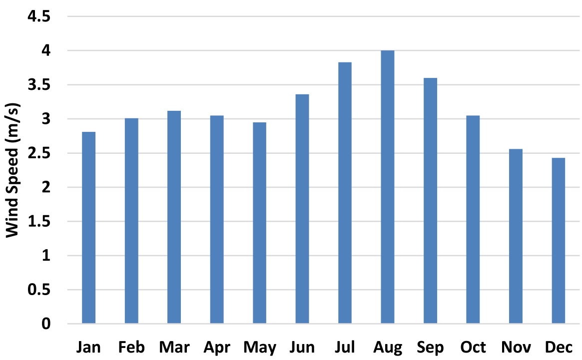

Fig. 6. Average monthly wind speed for the selected communities.

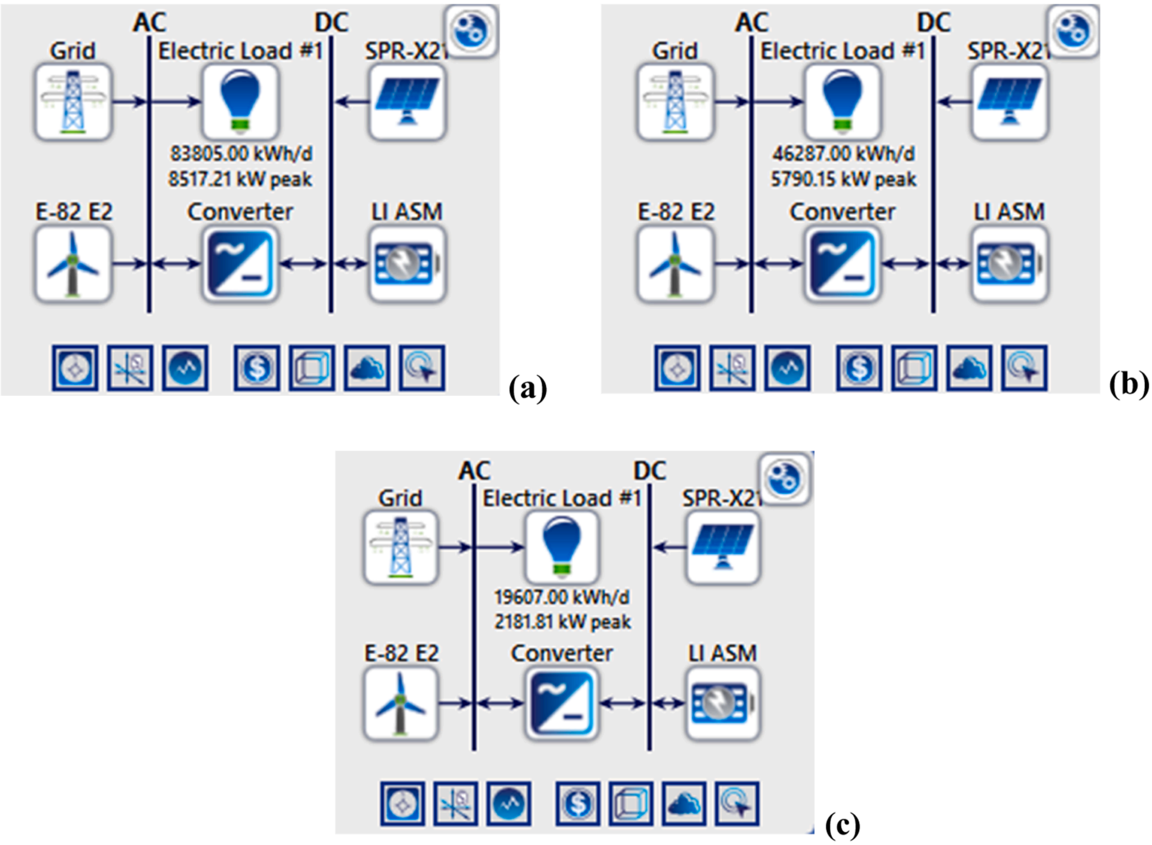

Fig. 7. On-grid hybrid system configuration for a) Aluu, b) Choba and c) Rumuekini.

operation and maintenance cost of $10 [64].

Grid

The price of electricity for residential households for service Band D consumers in March 2023 is 0.08625 $/kWh, while the price for businesses is 0.12052 $/kWh (see Table 2). Accordingly, this study adopts an average grid electricity rate of 0.08625 $/kWh (about N39.67/kWh) for its analysis. The excess electricity generated will be sold back to the grid at 10 % discount. For the purposes of this study, the currency rate of 1 $ is considered N459.94 on March 1, 2023 [17].

Economic modelling

The role of economic modelling in HOMER is crucial for evaluating the financial viability and cost-effectiveness of different energy generation system configurations. By quantifying the costs and benefits of different system configurations, economic modelling helps stakeholders make informed investments in clean and sustainable energy solutions.

Net present cost

Net present cost (NPC) is a financial metric used to evaluate the cost-effectiveness of an investment or project over its lifetime. It takes into account the time value of money by discounting future cash flows to their present value. NPC represents the total cost of a project, including initial investment, operating expenses, and any potential revenues or savings, all expressed in present value terms. The total NPC is calculated using the following expression [11]:

$$NPC = \frac{C_{ann,total}}{CRF(i,\ R_{proj})} \tag{1}$$

where $C_{ann.tot}$ is the annual total cost and *CRF* is capital recovery factor, *i* is interest rate in%, R_{proj} is project lifetime in years. The capital recovery factor is a ratio used to calculate the present value of an annuity (a series of equal annual cash flows). The value of the *CRF* is computed using the following equation [1]:

$$CRF\ (i.\ N) = \frac{i(1+i)^N}{(1+i)^N - 1} \tag{2}$$

where N refers to the number of years and i is calculated using [67]:

$$i = \frac{i_o - f}{1+f} \tag{3}$$

where, i_o is the nominal interest rate and f is the annual inflation rate.

Levelized cost of energy

Levelized cost of energy (LCOE) is calculated by dividing the total costs associated with the project (including capital costs, operational expenses, and fuel costs) by the total electricity generated over the project's lifetime. It represents the average cost per unit of electricity produced and allows for the comparison of different energy sources or technologies. It enables decision-makers to assess the cost competitiveness and economic viability of various energy generation options. The LCOE equation is presented in Eq. (4) [27]:

$$LCOE = \frac{C_{ann.tot}}{E_{prim.AC} + E_{prim.DC} + E_{grid.sales}} \tag{4}$$

where is $C_{ann.tot}$ is the annual total cost, $E_{prim.AC}$ is the AC primary load served, $E_{prim.DC}$ is the DC primary load served and $E_{grid.sales}$ is the total grid sales.

Return on investment

The Return on Investment (ROI) represents the net profit generated from an investment compared to the initial cost of investment. HOMER calculates the ROI with the following equation:

$$ROI = \frac{\sum_{i=0}^{R_{proj}} C_{i,ref} - C_i}{R_{proj}\left(C_{cap} - C_{cap,ref}\right)} \tag{5}$$

where $C_{i,ref}$ is the nominal annual cash flow for base (reference) system, C_i is the nominal annual cash flow for current system, R_{proj} is the project lifetime in years, C_{cap} is the capital cost of the current system and $C_{cap,ref}$ is the capital cost of the base (reference) system.

Simple payback period

The simple payback period is a critical factor in the economic analysis of systems, used to determine the duration required for an investment to recoup its initial costs through generated savings or revenues.

$$Simple\ Payback = \frac{Initial\ Investment}{Annual\ cash\ Inflows} \tag{6}$$

where the initial Investment is the total cost of the investment and the annual cash inflows represent the cash generated by the investment on an annual basis.

Discounted payback

Discounted payback is used to evaluate the time it takes for an investment to recoup its initial cost, considering the time value of money by discounting future cash flows. Unlike the Simple Payback Period, which doesn't account for the time value of money, the Discounted Payback Period takes into consideration the present value of future cash flows.

$$Discounted\ Payback\ Period = Number\ of\ Years\ before\ Full\ Recovery + \frac{Initial\ Investment - Cumulative\ Discounted\ Cash\ Flows\ at\ the\ End\ of\ the\ Last\ Full\ Year}{Cash\ Flow\ in\ the\ first\ year\ After\ Full\ Recovery} \tag{7}$$

Where number of years before full recovery represents the number of years before the cumulative discounted cash flows equal or exceed the initial investment, initial investment is the total cost of the investment, cumulative discounted cash flows at the end of the last full year is the sum of discounted cash flows up to the last full year before the investment is fully recovered, and cash flow in the first year after full recovery is the discounted cash flow in the year immediately following full recovery.

Internal rate of return (IRR)

Internal rate of return (IRR) is the discount rate at which the net present value (NPV) of all cash flows from the investment equals zero. In other words, it is the rate of return at which the present value of cash inflows equals the present value of cash outflows. The IRR is calculated as by using the following equation:

$$NPV = \sum_{t=0}^{n} \frac{CF_t}{(1+IRR)^t} = 0 \tag{8}$$

where CF_t the cash is flow at time t and n is the number of time periods and t is the time period.

Renewable fraction

Renewable fraction (RF) is a metric used to quantify the proportion of energy generated from renewable sources relative to the total energy produced within the system. It is dimensionless and calculated by Eq. (9) [45]:

$$RF = 1 - \frac{E_{nr} + H_{nr}}{E_s + H_s} \tag{9}$$

where E_{nr} is the non-renewable electrical production (kWh/yr), H_{nr} is the non-renewable thermal production (kWh/yr), E_s is the total electrical load served (kWh/yr), H_s is the total thermal load served (kWh/yr).

Homer calculations

Photovoltaic modules

The output power (P_{PV}) generated by the photovoltaic modules has been obtained from Wankouo Ngouleu et al. [71] and El-Houari at al. [25] as follows:

$$P_{PV} = P_{Rc} f_{PV} \left(\frac{G_T}{G_o}\right)[1 + \beta(T_c - T_o)] \tag{10}$$

where P_{Rc} is the rated capacity of the PV panel under standard test conditions in kW, f_{PV} is the PV derating factor [%], G_T is the solar radiation incident on the PV array in the current time step [kW/m^2], G_o is the incident radiation under standard test conditions [1 kW/m^2], β is the temperature coefficient of power [%/ °C], T_c is the PV cell temperature in the current time step [°C] and T_o is the PV cell temperature under standard test conditions [25 °C].

Wind turbine

HOMER employs a three-step process to compute the power output of the wind turbine in each time step.

First, the wind speed at the hub height of the wind turbine is calculated using Eq. (11) (the logarithmic law):

$$U_{hub} = U_{anem} \frac{In\left(\frac{z_{hub}}{z_o}\right)}{In\left(\frac{z_{anem}}{z_o}\right)} \tag{11}$$

where U_{hub} is the wind speed at the hub height of the wind turbine [m/s], U_{anem} is the wind speed at the hub height of the anemometer [m/s], z_{hub} is the hub height of the wind turbine [m], z_{anem} is the anemometer height [m] , z_o is the surface roughness length [m]. In a case the power law is chosen, wind speed at the hub height is calculated using the following equation, $\propto$ is the power law exponent:

$$U_{hub} = U_{anem} \left(\frac{z_{hub}}{z_{anem}}\right)^{\alpha} \tag{12}$$

After the hub height wind speed is determined, it utilizes the power curve of the wind turbine to calculate the expected power output from the wind turbine at that wind speed under standard conditions of temperature and pressure.

Finally, HOMER adjusts that power output value for the actual air density. The power output value for the actual air density is determined using Eq. (13):

$$P_{WT} = \left(\frac{\rho_a}{\rho_o}\right) P_o \tag{13}$$

where ρ_a denotes the actual density of air [kg/m^3], ρ_o denotes the air density at standard temperature and pressure and P_o denotes the wind turbine power output at standard temperature and pressure [kW].

Battery

The hourly battery energy during charge/discharge processes is determined using Eq. (14) [69]:

$$B_c = \frac{(E_{Load,\ d} \times D_A)}{\beta_{INV} \times \beta_{VRFB} \times DOD} \tag{14}$$

Where B_c is the battery capacity [kWh], $E_{Load,\ d}$ is the daily load consumption [kWh/day], D_A is the daily autonomy, DOD is the depth of discharge, β_{INV} is the inverter efficiency [%], and β_{VRFB} is the battery efficiency [%]

Methods of MCDM

To make a subjective, user-personalized ranking of the selection of renewable energy system, this research combines CRITIC and PROMETHEE II. The CRITIC methods are used to estimate the weights in determining the weights allocated to the criteria. The PROMETHEE II are used to determine the optimal alternative for the case study. The following subsection will focus on illustrating CRITIC and PROMETHEE II.

CRITIC

The CRITIC method developed by Diakoulaki et al. [23], is one of the popular objective weighing methods in the field of MCDM. The governing principle behind CRITIC uses standard deviation to measure contrast intensity of each criterion. The method ensures that a criterion with a higher contrast intensity or standard deviation is assigned with a higher weight. The basic process of CRITIC method is as described by the following steps [18]:

Step 1: A decision matrix X containing the criteria and alternatives related to the decision problem is created. Consider a decision matrix $X = [x_{ij}]_{m\times n}$, consisting of m alternatives and n criteria where $i = 1,\ldots,$ m and $j = 1,\ldots,$ n.

$$X = [x_{ij}]_{m\times n} = \begin{bmatrix} x_{11} & x_{12} & \ldots & x_{1n} \\ x_{21} & x_{22} & \ldots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \ldots & x_{mn} \end{bmatrix} \tag{15}$$

where x_{ij} represents the performance of measure of the i_{th} for the j_{th} criterion.

Step 2: The decision matrix r_{ij} is normalized using Eqs. (16) and (17) for beneficial criteria and non-beneficial criteria, respectively:

$$r_{ij} = \frac{x_{ij} - x_j^{min}}{x_j^{m,ax} - x_j^{min}};\ (i=1,2,\ldots,m; j=1,\ 2,\ldots,\ n) \tag{16}$$

$$r_{ij} = \frac{x_j^{max} - x_{ij}}{x_j^{m,ax} - x_j^{min}}\ (i=1,2,\ldots,m; j=1,\ 2,\ldots,\ n) \tag{17}$$

where x_j^{max} corresponds to the best performance value for jth criterion, and x_j^{min} is the worst performance value for jth criterion.

Step 3: Based on the value r_{ij} a vector of criteria was formed, each vector has a standard deviation σ_j, which represents the degree of deviation of alternatives for a given criterion. The standard deviation σ_j for each criteria is calculated using Eq. (18):

$$\sigma_{j=}\sqrt{\frac{1}{m}\left(\sum_{i=1}^{m} r_{ij} - \bar{r}_j\right)^2};\ j = 1,\ 2,\ldots,\ n \tag{18}$$

where m is a number of elements and $\bar{r}$ is an arithmetic mean.

Step 4: A symmetric matrix is conducted with dimension n x n that represents the linear correlation coefficient between r_j, r_k, which is noted as r_{jk}

$$r_{jk=}\frac{\sum_{i=1}^{m}(r_{ij} - \bar{r}_j)(r_{ik} - \bar{r}_k)}{\sqrt{\sum_{i=1}^{m}(r_{ij} - \bar{r}_j)^2 \sum_{i=1}^{m}(r_{ik} - \bar{r}_k)^2}};\ j,\ k = 1,2,\ \ldots,n \tag{19}$$

Step 5: The measure of the conflict created by criterion j with respect to the decision situation defined by the rest of criteria is calculated using Eq. (20):

$$R = \sum_{k=1}^{m} 1 - r_{jk};\ j = 1,\ 2, \ldots, n \tag{20}$$

Step 6: The quantity of the information in relation to each criterion is established using Eq. (21):

$$C_J = \sigma_j \sum_{k=1}^{m} \left(1 - r_{jk}\right) \tag{21}$$

Step 7: The value of C_J is normalized using Eq. (22) to determine the objective weight

$$w_j = \frac{C_J}{\sum_{k=1}^{n} C_k};\ k = 1,\ 2, \ldots, n \tag{22}$$

This objective weight will be used in PROMETHEE II to calculate alternatives' performance

PROMETHEE II

The PROMETHEE method was developed by Brans et al. [16] as an MCDM approach. PROMETHEE II is based on the concept of outranking relation, which involves comparing all pairs of alternatives to establish preferences and rankings. It considers the net outranking flows of each alternative to determine their overall preference. The method integrates preferences and indifferences to provide a complete ranking of actions. The stepwise procedure to solve the MCDM problem by PROMETHEE II are demonstrated below [20,39,40,74]:

Step 1: Normalize the decision matrix:

$$s_{ij} = \frac{x_{ij} - x_j^{min}}{x_j^{m,ax} - x_j^{min}};\ (i = 1, 2, \ldots, m; j = 1,\ 2, \ldots,\ n) \tag{23}$$

$$s_{ij} = \frac{x_j^{max} - x_{ij}}{x_j^{m,ax} - x_j^{min}}\ (i = 1, 2, \ldots, m; j = 1,\ 2, \ldots,\ n) \tag{24}$$

Step 2: Calculate the evaluative differences of the i^{th} alternative with respect to other alternatives.
Step 3: Calculate the preference function, $P_j(a,b)$. The preference function defines the degree by which a alternative is preferred over b alternative for j^{th} criterion. Now the preference function, $P_j(a,b)$. is computed using Eq. (24):

$$P_j(a,b) = 0,\ if\ R_{aj} \le R_{bj}\ and\ P_j(a,b) = \left(R_{aj} - R_{bj}\right)\ if\ R_{aj} > R_{bj} \tag{25}$$

Step 4: Calculate the aggregated preference function, $\pi(a,b)$

$$\pi(a,b) = \frac{\sum_{j=1}^{n} w_j P_j(a,b)}{\sum_{j=1}^{n} w_j} \tag{26}$$

Step 5: Determine the leaving (positive), $\varphi^{+}(a)$ and the entering (negative) outranking flows, $\varphi^{-}(a)$

$$\begin{aligned} \varphi^{+}(a) &= \frac{1}{m-1} \sum_{b=1}^{m} \pi(a,b)\ (a \ne b) \\ \varphi^{-}(a) &= \frac{1}{m-1} \sum_{b=1}^{m} \pi(b,a)\ (a \ne b) \end{aligned} \tag{27}$$

Step 6: Calculate the net outranking flow for each alternative

$$\varphi(a) = \varphi^{+}(a) - \varphi^{-}(a) \tag{28}$$

The alternative with highest value of $\varphi(a)$ outranks the other alternatives.

Assumptions and limitations in this study

In this analysis, economic conditions such as the exchange rate, discount rates, and inflation rates are assumed to remain constant due to software limitations. Specifically, a nominal interest rate of 11.5 % and an inflation rate of 15.7 % were employed, reflecting the prevailing economic landscape in Nigeria. This study did not factor in unexpected failures or maintenance requirements into the lifetime estimates for any system component; instead, reliance was placed on manufacturer specifications or industry standards. HOMER Pro also presupposes a certain level of grid reliability and availability, which may not fully capture the potential for unexpected outages or disruptions. Furthermore, the methodology does not account for intra-hour load variations, as the injection substation department operates solely on an hourly basis, which may not accurately capture the actual, more volatile demand patterns. Notwithstanding these caveats, HOMER Pro is still capable of generating a number of comprehensive theoretical conclusions that closely approximate real-world scenarios. Its broad usage and acceptance in comparable analyses worldwide is evidence of this.

Results and discussion

Electricity situation

Fig. 8 illustrates the graph of grid network electrical outages in the study area, based on data collected in 2022. Power outages are represented by black in this graphic, while periods of grid supply are highlighted in colour. The areas of Aluu, Choba, and Rumuekini are categorized as Band D users. According to their classification, Band D customers are entitled to a guaranteed minimum of 8 h of uninterrupted power supply daily. However, this standard is not being met. The distribution of electricity hours, is outlined in Table 3. The sheer amount of hours without access to electricity demonstrates unequivocally that these communities experience daily power disruptions. The existing connections to the national grid in these communities have a low capacity compared to the demand, leading to overloading and challenges in maintaining the transmission network. Among the three communities, Rumuekini recorded the highest number of electrified hours at 2211 h/yr, whereas Aluu had the lowest at 1687 h/yr. Conversely, Aluu experienced the greatest duration of low voltage disruptions, totalling 586 h/yr, whereas Rumuekini recorded the shortest duration of 308 h/yr. The low voltage is largely due to meter overload, resulting in adverse effects such as diminished appliance functionality, flickering or dimming of lights, and potential damage to equipment, among other issues. This makes individuals opt for autonomous solutions, such as private gasoline/diesel powered electricity generators to supply their electricity needs in the event of outages or instances of low voltage. This situation is particularly distressing due to the exhaust fumes and noise produced by thousands of these generators. Moreover, the expense of fuel in the open market is prohibitive and beyond the means of many villagers, in addition to the occasional fuel scarcity. Of the 24 h spent in a day, the Aluu, Choba and Rumuekini community users only get to enjoy an average duration of 4.6, 5.4, 6.2 h/day of electricity supplied to them from the national grid. Energy consumed in 2022 amounted to 5436.4 MW, 3375.4 MW and 1626.1 MW for Aluu, Choba and Rumuekini respectively. Table 4 displays the distribution of hours of electricity from 2021 to 2022, indicating a rise in the cumulative number of hours without supply over the course of 2022 compared to 2021. Specifically, there was a decrease of

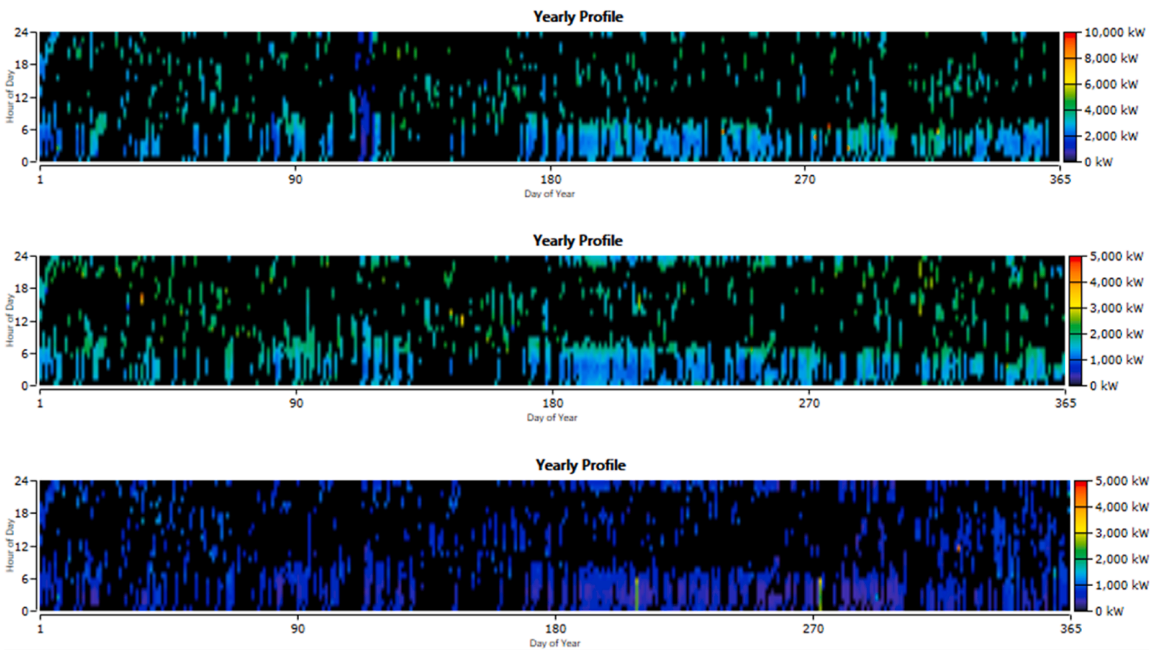

Fig. 8. Annual grid network outages in the study area for a) Aluu, b) Choba, and c) Rumuekini.

Table 3
Cumulative distribution of electricity (in hours) for 2022.

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Aluu													
NS	524	494	577	543	607	545	407	418	465	508	515	418	6021
LS	27	29	28	7	26	40	77	89	69	45	60	89	586
DS	0	0	0	22	0	0	11	0	0	0	0	0	33
SS	26	23	15	7	11	10	37	40	17	15	14	40	255
F	18	20	11	36	23	14	47	29	18	9	1	29	257
NH	148	104	113	164	77	110	189	167	151	167	130	167	1687
AE/D (MW)	444.6	347.2	368.1	419.9	290.5	368.0	601.55	506.5	497.8	581.9	503.7	506.6	5436.4
Choba													
NS	517	481	551	500	567	516	458	404	448	466	474	501	5883
LS	32	32	35	18	39	69	39	60	59	43	70	54	550
DS	0	0	0	0	0	0	0	0	0	0	0	0	0
SS	32	20	17	8	15	11	38	44	25	19	17	17	263
F	0	24	23	20	2	7	3	18	6	44	0	1	148
NH	163	115	118	174	119	117	206	194	182	172	159	171	1890
AE/D (MW)	300.9	227.3	225.6	311.9	215.2	214.1	323.1	323.1	299.9	306	309.3	319.2	3375.4
Rumuekini													
NS	505	480	552	448	518	539	374	409	448	464	501	446	5684
LS	30	12	18	8	16	24	30	67	62	15	10	16	308
DS	0	0	0	22	0	16	10	0	0	0	0	0	48
SS	25	26	16	8	9	12	34	43	26	22	16	21	258
F	2	8	19	97	2	21	10	0	6	10	0	0	175
NH	168	146	139	197	126	140	243	222	168	220	193	249	2211
AE/D (MW)	140.6	129.4	112.5	123.5	102.9	100.1	165.4	154.7	108.6	135.9	144.8	207.7	1626.1

NS: No supply; LS: Low supply; DS: Disconnected; SS: Service shutdown; F: Fault; NH: Number of electrified hours; AE/D: Total Amount of Energy Supplied in MW.

Table 4
Distribution of electricity (in hours), 2021 and 2022.

	2021			2022		
	Aluu	Choba	Rumuekini	Aluu	Choba	Rumuekini
NS	5263	5521	5504	6021	5883	5688
LS	399	416	415	586	550	308
DS	41	53	0	33	0	48
SS	444	419	386	255	263	258
EF	126	135	56	257	148	169
NH	2180	2187	2334	1687	1890	2212
AE/D (MW)	6549.5	3791.5	2002.5	5436.4	3375.4	1635.7

493, 297 and 122 h with supply across all communities during 2022 when contrasted with the previous year.

Results of HOMER pro

This section presents the results and discussion of the renewable energy system used to fulfil the needs of three selected communities in Rivers State, Nigeria. The selection of the optimal system in HOMER is predicated upon minimizing NPC and LCOE. The outcomes of the HOMER simulations identified five best-performing system configurations: 1) grid/PV (C_1), 2) grid/PV/Wind (C_2), 3) grid/Wind (C_3), 4) grid/PV/Battery (C_4), and 5) grid/Wind/Battery (C_5). These configurations, which originate from the assessed potential system combinations detailed in Table 5, are systematically organized based on their technical, economic, and environmental performance. For Aluu, the NPC, COE, operating cost and RF are $41.8million–$55.4million, 0.0181–0.0218$/kWh, 5829–27,984 $/yr, and 68.2 %–77.3 % respectively. In Choba, the NPC, LCOE, operating cost and RF are $23.0 million–$27.8 million, 0.0183–0.0217 $/kWh, 2491–90,982 $/yr, and 68.4 %–76.4 % respectively. For Rumuekini, the NPC, COE, operating cost, and RF are $9.87 million–$13.5 million, 0.0185–0.0234 $/kWh, 5044–29,511 $/yr and 68.0 %–76.0 % respectively. The table further illustrates that as the renewable fraction of hybrid renewable systems increases, both the NPC and the COE increase accordingly.

Based on the data presented in Table 5, several key observations can be made. For Aluu, C_5 emerges as the energy option with the highest levels of renewable fraction and the lowest annual CO_2 emissions making it the most environmentally friendly choice. Meanwhile, C_1 distinguishes itself with the least LCOE of $0.0181/kWh and also offers the lowest NPC and initial capital cost. This renders C_1 particularly attractive for those prioritizing financial risk mitigation and ensuring a favourable return on investment. In terms of renewable energy production, C_4 leads with the highest renewable energy output at 61,583,965 kWh/year; however, it produces an excess of 3943,698 kWh/year. Meanwhile, C_2 is characterized by the lowest operating cost. Notably, C_3 and C_5 exhibit zero excess energy, indicating a well-balanced energy system with minimal wastage. For Choba, C_1 emerges with the lowest operating

Table 5
Optimization results of the proposed system.

Item (Unit)	Aluu					Choba					Rumuekini				
	C_1	**C_2**	**C_3**	**C_4**	**C_5**	**C_1**	**C_2**	**C_3**	**C_4**	**C_5**	**C_1**	**C_2**	**C_3**	**C_4**	**C_5**
System Configuration															
PV (kW)	33,500	28,000		34,500		19,000	15,500		19,000		8000	–		9000	
Wind		4	26		27		2	12		13		–	6		6
Converter (kW)	18,500	16,000		17,500	1000	9500	8500		9500	500	4000	–		5000	136
Battery				4280	4280				2361	1180		–		1000	1000
Economic performance															
Initial capital cost ($)	41.4M	42.9M	52.0M	42.4M	55.1M	22.9M	23.1M	24.0M	23.3M	26.6M	9.66 M	–	12.0M	11.3M	12.2 M
NPC	41.8M	43.1M	53.2 M	42.7M	55.4M	23.0M	23.2M	27.8M	23.7M	27.5M	9.87M	–	12.6M	11.6M	13.5 M
Operating cost ($/yr)	10,295	5829	27,984	7513	6672	2491	2520	90,982	11,263	21,941	5044	–	14,014	7122	29,511
LCOE ($/kWh)	0.0181	0.0192	0.0215	0.0183	0.0218	0.0183	0.0189	0.0217	0.0187	0.0204	0.0185	–	0.0219	0.0194	0.0234
ROI (%)	2.2	1.9	0.5	2.1	0.3	2.2	2.1	1.2	2.1	0.9	2.2	–	0.9	1.3	0.3
1RR (%)	3.7	3.2	0.9	3.5	0.5	3.7	3.5	1.9	3.5	1.5	3.7	–	1.7	2.2	0.5
Simple Payback (yr)	18.09	18.05	18.13	18.41	18.99	17.97	17.65	16.18	18.39	17.25	17.98	–	22.21	21.06	24.91
Discounted Payback (yr)	11.45	11.74	13.67	11.66	14.04	11.52	11.55	12.49	11.67	12.94	11.52	–	13.52	12.87	14.08
Technical performance															
Renewable energy production (kWh/yr)	60,376,828	57,691,821	59,075,698	61,583,965	60,606,509	33,784,776	31,714,470	30,624,961	33,783,141	32,269,712	14,298,615	–	13,710,155	15,515,770	13,709,433
Excess electricity (kWh/yr)	3192,511	2297,789	0	3943,698	0	2612,673	1520,229	0	2293,889	0	1099,759	–	0	712,590	0
Unmet electrical load (kWh/yr)	0	0	0	0	0	0	0	0	0	0	0	–	0	0	0
Capacity shortage (kWh/yr)	0	0	0	0	0	0	0	0	0	0	0	–	0	0	0
Environmental performance															
Renewable fraction (%)	68.2	72.2	76.4	68.5	77.3	68.4	72.3	74.3	68.7	76.4	68.0	–	76	71.8	76.0
CO_2 emissions (kg/yr)	11,094,190	9451,881	8809,148	11,049,222	8679,440	6002,862	5120,143	4979,137	6001,829	4820,633	2574,317	–	2081,735	2535,759	2081,279

Table 6
Decision matrix.

		Aluu					Choba					Rumuekini			
Criteria	Symbol	C_1	C_2	C_3	C_4	C_5	C_1	C_2	C_3	C_4	C_5	C_1	C_3	C_4	C_5
Initial capital cost ($)	C_{E-1}	41.4M	42.9M	52.0M	42.4M	55.1M	22.9M	23.1M	24.0M	23.3M	26.6M	9.66 M	12.0M	11.3M	12.2 M
NPC	C_{E-2}	41.8M	43.1M	53.2 M	42.7M	55.4M	23.0M	23.2M	27.8M	23.7M	27.5M	9.87M	12.6M	11.6M	13.5 M
Operating cost ($/yr)	C_{E-3}	10,295	5829	27,984	7513	6672	2491	2520	90,982	11,263	21,941	5044	14,014	7122	29,511
LCOE ($/kWh)	C_{E-4}	0.0181	0.0192	0.0215	0.0183	0.0218	0.0183	0.0189	0.0217	0.0187	0.0204	0.0185	0.0219	0.0194	0.0234
ROI (%)	C_{E-5}	2.2	1.9	0.5	2.1	0.3	2.2	2.1	1.2	2.1	0.9	2.2	0.9	1.3	0.3
Discounted Payback (yr)	C_{E-6}	11.45	11.74	13.67	11.66	14.04	11.52	11.55	12.49	11.67	12.94	11.52	13.52	12.87	14.08
Renewable energy production (kWh/yr)	C_{T-1}	60,376,828	57,691,821	59,075,698	61,583,965	60,606,509	33,784,776	31,714,470	30,624,961	33,783,141	32,269,712	14,298,615	13,710,155	15,515,770	13,709,433
Excess electricity (kWh/yr)	C_{T-2}	3192,511	2297,789	0	3943,698	0	2612,673	1520,229	0	2293,889	0	1099,759	0	712,590	0
Renewable fraction (%)	$C_{\in-1}$	68.2	72.2	76.4	68.5	77.3	68.4	72.3	74.3	68.7	76.4	68.0	76	71.8	76.0
CO_2 emissions (kg/yr)	$C_{\in-2}$	11,094,190	9451,881	8809,148	11,049,222	8679,440	6002,862	5120,143	4979,137	6001,829	4820,633	2574,317	2081,735	2535,759	2081,279
Technology readiness	C_{s-1}	4.1	3.7	1.3	3.4	1.2	4.1	3.7	1.3	3.4	1.2	4.1	1.3	3.4	1.2
Ease of installation	C_{s-2}	4.4	3.9	1.2	4.3	1.1	4.4	3.9	1.2	4.3	1.1	4.4	1.2	4.3	1.1
Land use	C_{s-3}	2.8	2.5	2.2	2.9	2.3	2.8	2.5	2.2	2.9	2.3	2.8	2.2	2.9	2.3

1Strongly Agree.
2 Disagree.
3 Neutral.
4 Agree.
5 Strongly Agree.

cost, initial capital, NPC and LCOE, positioning it as an economically sound investment opportunity. With reduced financial risk, improved cash flow and improved profitability potential, C_1 stands out as a highly desirable choice. Additional it has the highest renewable energy production of 33,784,776 kWh/year, though it also has the highest excess energy. Conversely, C_5 features the highest RF, establishing it as the most environmentally friendly energy option. For Rumuekini, C_1 exhibits the lowest operating cost, initial capital cost, and NPC, rendering it the most economical option and an attractive investment opportunity, however, it also generates the highest excess energy of 1099,759 kWh/year. In terms of energy production, C_4 demonstrates the highest output of renewable energy, generating 15,515,770 kWh per year. Additionally, C_5 offers the highest levels of RF and has the lowest annual CO_2 emissions, establishing it as the most environmentally friendly energy alternative. Furthermore, C_2 is deemed unfeasible for this town. Balancing these criteria, alongside social considerations, is crucial for attaining optimal outcomes and ensuring the project's long-term success and sustainability. To address this challenge, the use of MCDM techniques becomes indispensable in identifying the optimal configuration of the hybrid system. The findings from the MCDM analysis will be detailed in the following section.

Results of multi-criteria decision-making

HOMER proves to be a valuable tool for optimizing hybrid microgrid and grid-connected system design and analysis. Integrating MCDM techniques can greatly augment its efficacy, offering a more organized and thorough approach to decision-making, particularly in scenarios involving multiple conflicting objectives and stakeholder preferences. This section identifies the optimal hybrid system choice by considering multiple criteria across technical, economic, environmental, and social dimensions, each with various attributes. The initial decision matrix, shown in Table 6, evaluates five scenarios against thirteen criteria. The values assigned to these attributes are derived from HOMER simulations (see Table 5) and expert opinions. Expert opinions were gathered through a survey administered using a 5-point Likert scale. The CRITIC-PROMETHEE II approach was used to determine the optimum scenario. From the thirteen criteria, initial capital cost (C_{E-1}), NPC (C_{E-2}), operating cost (C_{E-3}), LCOE (C_{E-4}), ROI (C_{E-5}) and discounted payback (C_{E-6}) are economic criteria, renewable energy production (C_{T-1}) and excess electricity (C_{T-2}) are technical criteria, renewable fraction ($C_{\in-1}$) and CO_2 emissions ($C_{\in-2}$) are environmental criteria and technology readiness (C_{s-1}), ease of installation (C_{s-2}) and land use (C_{s-3}) are social criteria.

CRITIC results

The CRITIC method was utilized to determine the weight of each criterion. Among the criteria considered— C_{E-1}, C_{E-2}, C_{E-3}, C_{E-4}, C_{E-6}, C_{T-2}, and $C_{\in-2}$—classified as non-beneficial, lower minimum values were preferred. Conversely, for C_{E-5}, C_{T-1}, $C_{\in-1}$, C_{s-1}, C_{s-2} and C_{s-3}, identified as beneficial criteria, higher minimum values were preferred. The weights assigned to each criterion via the CRITIC approach are presented in Table 7. According to the results, Aluu and Choba received the highest weightage of 0.202 and 0.186 for technology readiness (C_{s-1}) and for Rumuekini, CO_2 emissions ($C_{\in-2}$) received the highest weightage of 0.208. Aluu and Choba had the lowest weights for initial capital cost (C_{E-1}), with values of 0.0998 and 0.0995 respectively, while Rumuekini's NPC (C_{E-2}) was assigned the least weight of 0.0917. Fig. 9 illustrates the weights assigned to economic, environmental, technical, and social performances. Notably, economic performance has the highest weight, followed by social performance, then technical performance, and finally environmental performance. However, for Aluu, technical performance has the lowest weight (Table 7).

PROMETHEE II results

Table 8 provides a comprehensive summary of the scenario rankings based on their net outranking scores. For Aluu and Choba, the highest-ranking option is C_1, signifying the grid/PV combination, with a net score of 0.268 and 0.276 respectively. Conversely, the least favourable option is C_3, which represents the grid/Wind combination, with a net score of −0.356 and −0.376 for Aluu and Choba, respectively. Similarly, in Rumuekini, option C_1, indicative of the grid/PV configuration, achieved the foremost rank with a net score of

Table 7
Weights of the performance criteria.

Criteria	Aluu		Choba		Rumuekini	
	Weight	**Rank**	**Weight**	**Rank**	**Weight**	**Rank**
C_{E-1}	0.099803505	13	0.099535678	13	0.098066093	11
C_{E-2}	0.102928651	11	0.114376463	8	0.09174527	13
C_{E-3}	0.105009983	10	0.104036943	11	0.102562622	10
C_{E-4}	0.106581649	8	0.099598138	12	0.103352025	9
C_{E-5}	0.196639364	2	0.170645361	3	0.16828547	6
C_{E-6}	0.102760891	12	0.104393016	10	0.091945394	12
C_{T-1}	0.106069824	9	0.139117098	7	0.152196007	7
C_{T-2}	0.180206132	5	0.168297303	4	0.19184712	4
$C_{\in-1}$	0.109762382	7	0.110575397	9	0.107331553	8
$C_{\in-2}$	0.181785899	4	0.160443921	5	0.207962667	1
C_{s-1}	0.19610042	3	0.175950084	2	0.194296006	3
C_{s-2}	0.202611145	1	0.1856726	1	0.204730361	2
C_{s-3}	0.164122697	6	0.14857428	6	0.179725208	5

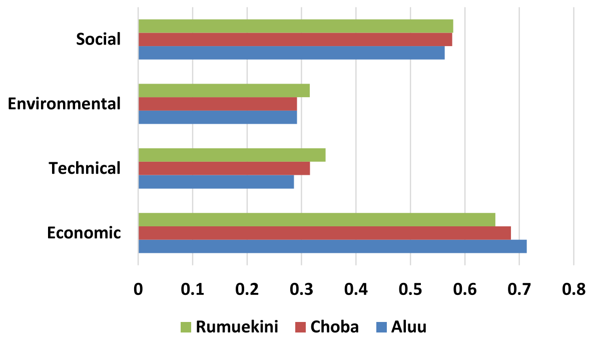

Fig. 9. Weights of the economic, environment and social performances.

Table 8
Leaving Flow, entering flow and ranking of all the considered feasible configurations.

Case study	Leaving Flow, $\varphi^+(a)$	Entering flow, $\varphi^-(a)$	$\varphi(a)$	Rank
Aluu				
C_1	0.373629623	0.105832146	0.267797477	1
C_2	0.299602161	0.164464174	0.135137987	3
C_3	0.138012967	0.493815240	−0.355802273	4
C_4	0.364434803	0.106084148	0.258350655	2
C_5	0.146137174	0.451621019	−0.305483845	5
Choba				
C_1	0.382530099	0.10687498	0.275655119	1
C_2	0.308689546	0.112868164	0.195821382	3
C_3	0.139098144	0.51469229	−0.375594147	5
C_4	0.323065659	0.091226359	0.231839299	2
C_5	0.136674442	0.464396095	−0.327721653	4
Rumuekini				
C_1	0.49125037	0.184595763	0.306654607	1
C_3	0.172179901	0.283348290	−0.111168389	3
C_4	0.353379699	0.170517474	0.182862225	2
C_5	0.105384888	0.483733331	−0.378348443	4

0.307. In contrast, option C_5, which denotes the grid/Wind/Battery configuration, was the least favourable, scoring −0.378.

Performance evaluation of optimum case study

For Aluu

The optimal solution for providing electricity to the Aluu community, considering the grid's unreliable power supply, identifies C_1 as the superior choice. Table 5 shows the system architecture without wind turbines or battery storage units: 33,500 kW of PV modules and a necessary 18,500 kW system converter. This system achieves the minimum NPC and the lowest LCOE at $0.0181, outperforming alternative configurations such as C_2, C_3, C_4, and C_5, which exhibit higher LCOEs of $0.0192, $0.0215, $0.0183 and $0.0218, respectively. Additionally, the system incurs an annual operating cost of $10,295, attains an RF of 68.2 %, and generates excess energy amounting to 3192,511 kWh% with no capacity shortage. Table 9 shows the annual energy production and consumption. It is clear that the grid provides 29.1 % of the electricity generated, with the PV modules accounting for 70.9 %. This indicates that a huge

Table 9
Annual electrical production and utilization of the optimized C_1 system in Aluu.

Production	kWh/yr	%
SunPower X21-335-BLK	42,822,730	70.9
Grid Purchases	17,554,098	29.1
Total	60,376,828	100
Consumption		
AC primary load	30,627,150	55.5
Grid sales	24,575,656	44.5
Total	55,202,806	100
Quantity		
Excess electricity	3192,511	5.29
Unmet load	0	0
Capacity shortage	0	0

percentage of electricity is obtained from the PV modules. The total power consumption from AC primary loads is 30,627,150 kWh/yr (55.5 %) and grid sales is 24,575,656 kWh/yr (44.5 %), so it can be concluded that the optimal hybrid system can meet this level of power consumption. The SunPower X21–335-BLK PV output, energy purchased from the grid and energy sold to the grid are depicted in Fig. 10. Among the five renewable energy system alternatives, the optimal system C_1 has the least discounted payback period of 11.45 years. A shorter payback period enhances the attractiveness of the investment by providing quicker returns, thereby reducing the time and financial risk associated with recouping the initial capital outlay. Table 10 illustrates the cost distribution of each system component under C_1 at Aluu, for the duration of the project lifecycle. In the table, positive values indicate expenditures (cash outflow), while negative values denote income (cash inflow). The total capital cost for this optimal stem is $41,397,000.00 with a total operation and maintenance cost of $8738,494.55 and total NPC of $41,828,171.59. Additionally, the annual expenditures for the components of this optimal hybrid renewable energy system is $998,708.75. The second best hybrid renewable energy system on the optimal energy system list is C_4 which is the combination of PV modules and battery storage units. During grid outages, C_4 can supply backup power from stored energy, enhancing reliability. This backup feature is advantageous, especially in areas like Aluu with inconsistent grid infrastructure. A very interesting choice is option C_3 which can be observed to have an NPC approximately 3 % higher than C_1 and the least operating cost of $5829/yr. The monthly average power production for system C_3 is illustrated in Fig. 11. It can be observed that the penetration of wind energy is lower, especially during December. The proposed framework choose the C_5 option as the least choice because it exhibited the highest NPC and LCOE, which are 24.5 % and 17.0 % higher than the C_1 system. The PV module technology is the most established renewable energy technology in Nigeria because of the ubiquity of solar radiation throughout the country. This extensive solar resource makes PV systems a highly practical and efficient solution for meeting energy demands across diverse sectors, including residential, commercial, and rural electrification projects. The locations considered for the proposed system are represented with red markers (Fig. 12a).

For Choba

According to the optimization results, the most efficient configuration of system C_1 achieves an RF of 68.4 %, utilizing 19,000 kW of PV modules and a 9500 kW system converter. This configuration results in the lowest annual operating costs of $2491 and an LCOE of $0.0183/kWh. This cost is 78.8 % lower than the current residential rate and 84.8 % lower than the commercial rate, demonstrating the economic feasibility of this design and the substantial savings it offers for all customers. Table 11 shows the annual energy production and consumption of the C_1 system. The power grid supplies 28.1 % of the total electricity generated, while PV modules contribute 71.9 %, indicating that a significant portion of electricity is sourced from solar energy. The energy consumption from AC primary load is 16,894,755 kWh/yr, which is 56.2 % of total utilization. Energy consumption from grid sales is 13,193,653 kWh/yr, which constitutes 42.8 % of total consumption. The total energy consumption is 30,088,408 kWh/year. The SunPower X21–335-BLK PV output, energy purchased from the grid and energy sold to the grid are depicted in Fig. 13. The annual excess electricity produced for the system amounts to 2612,673 kWh/yr, equalling only 7.73 % of the total generation, which falls within acceptable limits. Table 12 provides a comprehensive breakdown of the cost analysis, detailing that the total system capital cost, operation and maintenance cost, and NPC amount to $22,933,000.00, $4604,437.29, and $23,037,310.19, respectively. It is essential to highlight that the annual expenditures for the components of this optimal hybrid renewable energy system calculated was $550,049.46. The system is projected to achieve a discounted payback period of 11.52 yrs to recoup the investments, with an IRR of 3.7 %. Contrasting the optimal hybrid system architecture with C_3 (the least preferred option), a proportional reduction of approximately 17 % in emissions outputs is observed in comparison to option C_1. However, it is important to note that option C_3 incurs higher costs, with initial capital, NPC, and LCOE increasing by 4.6 %, 17.3 %, and 15.6 %, respectively, compared to C_1. Additionally, the operating costs for C_3 are significantly higher, at $90,982/kWh compared to $C_{1's}$ $2491/kWh. Furthermore, C_1 is preferred due to its higher indicators

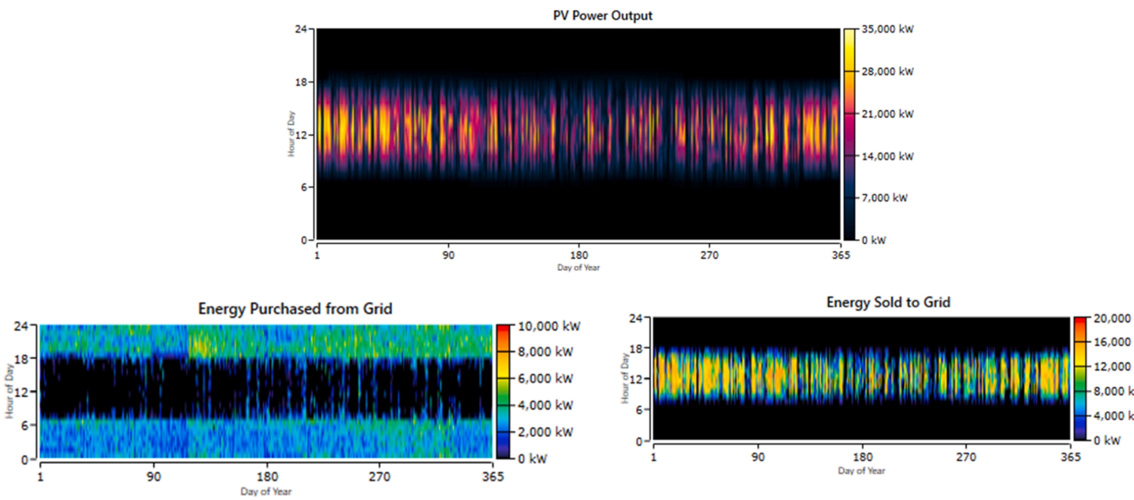

Fig. 10. PV module output, energy purchased from the grid and sold to the grid (Aluu).

Table 10
Cost summary of the optimized C_1 system in Aluu.

Component	Capital ($)	Replacement ($)	O&M ($)	Salvage Cost ($)	Total ($)
Total NPC					
Grid	0.00	0.00	−16,486,711.14	0.00	−16,486,711.14
SunPower X21-335-BLK	31,222,000.00	0.00	0.00	0.00	31,222,000.00
System Converter	10,175,000.00	17,717,881.45	7748,216.59	−8548,215.31	27,092,882.73
System	41,397,000.00	17,717,881.45	−8738,494.55	−8548,215.31	41,828,171.59
Annualized costs					
Grid	0.00	0.00	−393,644.33	0.00	−393,644.33
SunPower X21-335-BLK	745,470.90	0.00	0.00	0.00	745,470.90
System Converter	242,943.00	423,040.33	185,000.00	−204,101.14	646,882.19
System	988,413.90	423,040.33	−208,644.33	−204,101.14	998,708.75

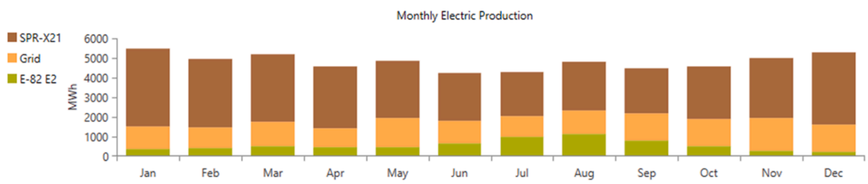

Fig. 11. Monthly average electrical production for system C_3.

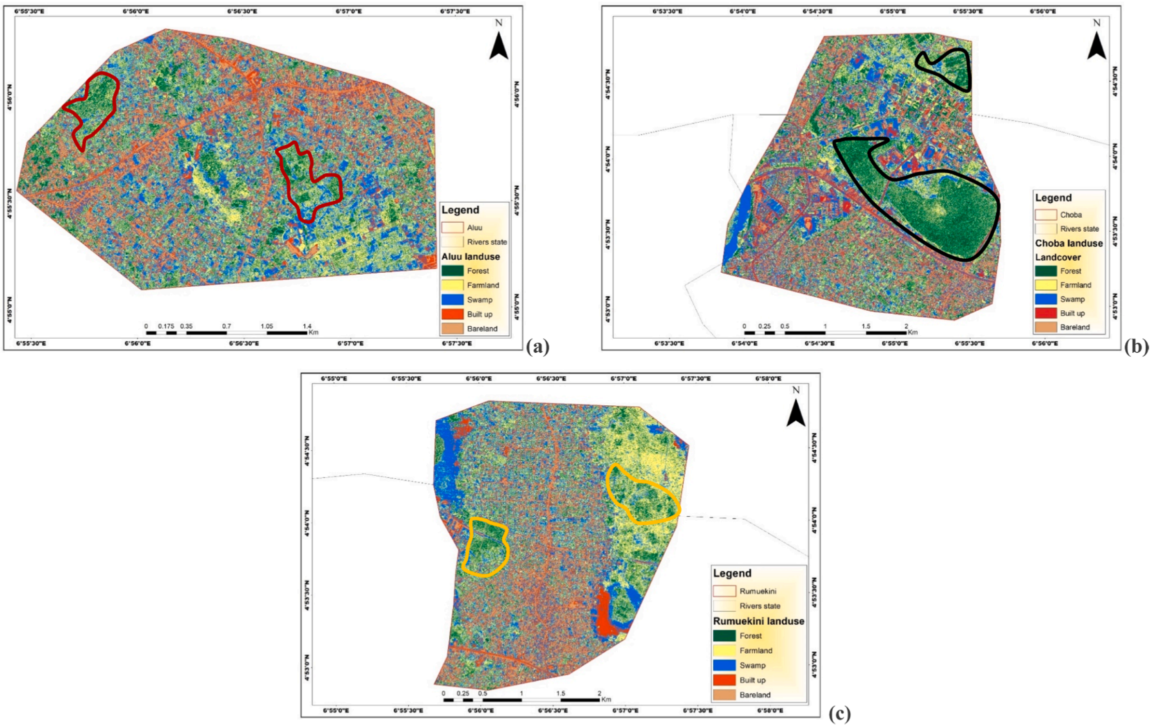

Fig. 12. Landsat image of the study area for March 2024 covering a) Aluu, b) Choba, and c) Rumuekini communities.

in technology readiness, ease of installation, and land use efficiency compared to C_3. The location under consideration for the proposed system are depicted by black markers in Fig. 12b.

For Rumuekini

The optimal solution for supplying the Rumuekini community from the grid includes an 8000 kW PV module system paired with a 4000 kW system converter. In this configuration, power is sourced from both the PV modules and grid purchases. This configuration has been identified as having the lowest NPC and LCOE, amounting to $9.66 million and $5044/kWh, respectively, in addition to its minimal operating costs. Table 13 shows the annual energy production and consumption of this system. It is clear that the power grid

Table 11
Annual electrical production and utilization of the optimized C_1 system in Choba.

Production	kWh/yr	%
SunPower X21–335-BLK	24,286,577	71.9
Grid Purchases	9498,199	28.1
Total	33,784,776	100
Consumption		
AC primary load	16,894,755	56.2
Grid sales	13,193,653	42.8
Total	30,088,408	100
Quantity		
Excess electricity	2612,673	7.73
Unmet load	0	0
Capacity shortage	0	0

Table 12
Cost summary of the optimized C_1 system in Choba.

Component	Capital ($)	Replacement ($)	O&M ($)	Salvage ($)	Total ($)
Total NPC					
Enercon E-82 E2 [2MW]	0.00	0.00	−8583,251.21	$0.00	−8583,251.21
Grid	17,708,000.00	0.00	0.00	$0.00	17,708,000.00
SunPower X21-335-BLK	5225,000.00	9098,371.55	3978,813.93	−4389,624.08	13,912,561.40
System Converter	22,933,000.00	9098,371.55	($4604,437.29)	−4389,624.08	23,037,310.19
Annualized costs					
Grid	0.00	0.00	−204,937.67	0.00	−204,937.67
SunPower X21-335-BLK	422,804.39	0.00	0.00	0.00	422,804.39
System Converter	124,754.51	217,236.92	95,000.00	−104,808.69	332,182.75
System	547,558.91	217,236.92	−109,937.67	−104,808.69	550,049.46

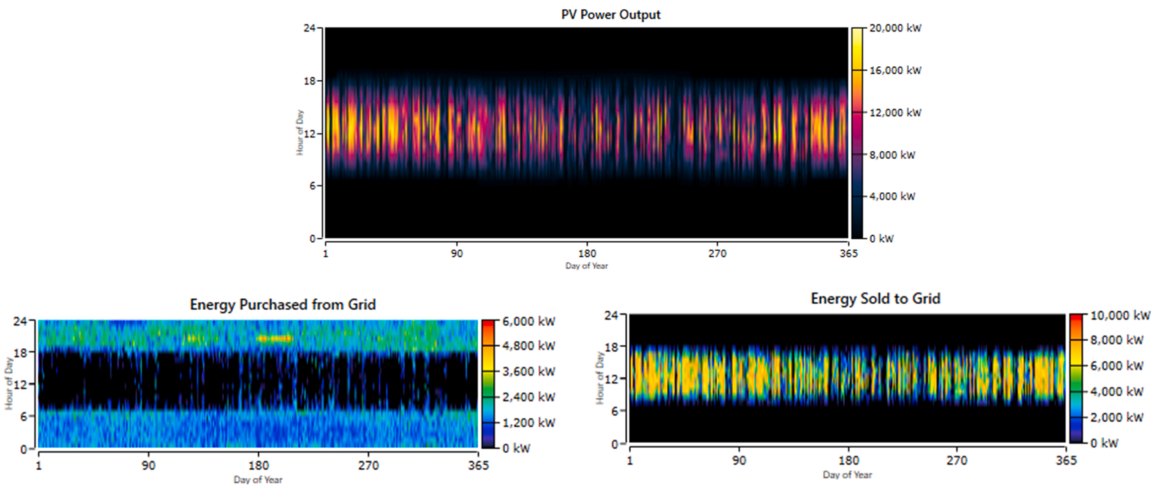

Fig. 13. PV module output, energy purchased from the grid and sold to the grid (Choba).

and PV modules provides 28.5 % and 71.5 % of the electricity generated respectively. The renewable energy fraction reaches 71.5 % because Rumuekini, similar to Aluu and Choba, benefits from high solar irradiance, which enables the PV modules to operate at peak efficiency, capturing and converting sunlight into electricity with minimal interruptions. This results in a steady and reliable output from the PV modules, making solar energy a dominant contributor to the overall energy mix. The total power consumption from AC primary loads is 7156,555 kWh/yr (52.2 %) and grid sales is 6,553,600 kWh/yr (47.8 %). The SunPower X21–335-BLK PV output, energy purchased from the grid and energy sold to the grid are depicted in Fig. 14. Table 14 illustrates the cost distribution of each system component in this configuration for the duration of the project lifecycle. The total system capital cost, total operation and maintenance cost and total NPC is given by $9656,000.00, $1771,364.92 and $9867,265.60 respectively. The annual expenditures for the components of this optimal hybrid renewable energy system total $235,595.39. The second most suitable hybrid renewable energy system on the optimal energy system list is C_4, which consists of PV modules and battery storage units. This system achieves the highest renewable energy production, generating 15,515,770 kWh/year. However, its operating cost and NPC increase by 29 % and 14.9 %,

Table 13
Annual electrical production and utilization of the optimized C_1 system in Rumuekini.

Production	kWh/yr	%
SunPower X21-335-BLK	10,225,327	71.5
Grid Purchases	4073,287	28.5
Total	14,298,615	100
Consumption		
AC primary load	7156,555	52.2
Grid sales	6553,600	47.8
Total	13,710,155	100
Quantity		
Excess electricity	0	0
Unmet load	0	0
Capacity shortage	0	0

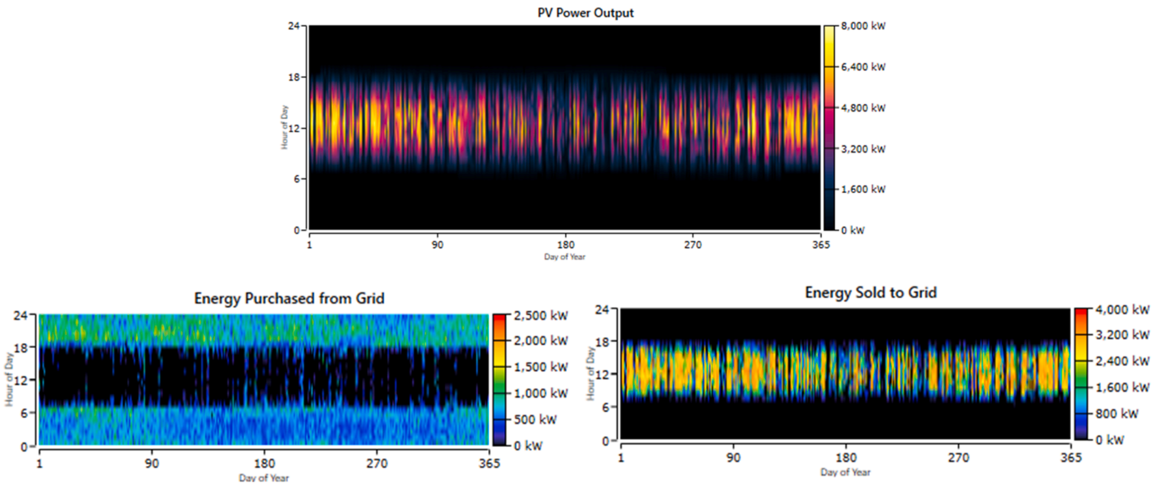

Fig. 14. PV module output, energy purchased from the grid and sold to the grid (Rumuekini).

Table 14
Cost summary of the optimized C_1 system in Rumuekini.

Component	Capital ($)	Replacement ($)	O&M ($)	Salvage ($)	Total ($)
Total NPC					
Grid	0.00	0.00	−3446,654.99	0.00	−3446,654.99
SunPower X21-335-BLK	7456,000.00	0.00	0.00	0.00	7456,000.00
System Converter	2200,000.00	3830,893.29	1675,290.07	−1848,262.77	5857,920.59
System	9656,000.00	$3830,893.29	−1771,364.92	−1848,262.77	9867,265.60
Annualized costs					
Grid	0.00	0.00	−82,293.93	0.00	−82,293.93
SunPower X21-335-BLK	178,022.90	0.00	0.00	0.00	178,022.90
System Converter	52,528.22	91,468.18	40,000.00	−44,129.98	139,866.42
System	230,551.12	91,468.18	-$42,293.93	−44,129.98	235,595.39

Table 15
Comparison between this study and other grid-connected systems.

Author	Location	System	LCOE ($/kWh)	RF (%)
[34]	Nigeria	Grid/PV/Wind/Biogas	0.0788	94.6
[32]	India	Grid/PV/Wind/Micro Hydro	0.034	91.5
[70]	Pakistan (Gujar, Awaran; Plantak, Washuk; Shahrak, Kech)	Grid/PV/Wind/Battery	−0.00412; −0.00523; −0.00721	–
[56]	Indonesia	Grid/PV/Wind/Battery	0.0446	82
[60]	Saudi Arabia	Grid/PV/Battery	0.117	48.6
[7]	Nigeria	Grid/PV	0.1904	–
Present Study	Nigeria (Aluu, Choba, Rumuekini)	Grid/PV	0.0181–0.0183	68–68.4

respectively, compared to the operating cost and NPC of C_1. The proposed framework identifies the C_5 option as the least recommended due to its significantly higher costs. It has an NPC that is 10.1 % higher, an LCOE that is 22.6 % higher, and operating costs that are more than five times greater than the C_1 system. The location under consideration for the proposed system are depicted by yellow markers in Fig. 12c.

Comparing the proposed design with other grid-connected systems

Table 15 provides a comparative analysis of the results from the proposed system against earlier studies involving grid-connected systems. Due to the complexities in comparing the current study with existing literature—stemming from variations in system characteristics and climate conditions—the LCOE was selected as the primary metric for evaluation and comparison with analogous case studies. The LCOE for the proposed system in this study is notably low compared to most grid-connected studies, except for those by [70]. From this system analysis, it was deduced that grid connected PV system is competitive and at a mature stage in the country.

Sensitivity analysis

Sensitivity analysis is used to evaluate how changes in various input variables affect the outcomes of an energy system simulation. This type of analysis is essential for understanding the robustness of the energy system design under various scenarios and assumptions. In this section, some of the system's variables have been investigated by sensitivity analysis to see impact on key performance indicators, such as NPC and LCOE on the optimal scenario (C_1). The economic variables under consideration encompassed the PV cost, discount rate and inflation rate. The inflation rate holds pivotal significance in assessing the time value of money within cash flow

Table 16
Impact of a) PV cost b) discount rate c) c) inflation rate d) grid power price e) scaled solar irradiance and f) electrical demand on the NPC and LCOE.

	Aluu			Choba			Rumuekini	
Parameter	**NPC ($)**	**LCOE ($/kWh)**	**Parameter**	**NPC ($)**	**LCOE ($/kWh)**	**Parameter**	**NPC ($)**	**LCOE ($/kWh)**
PV cost ($)								
652.4	32.5	0.0142	**652.4**	17.7	0.0141	**652.4**	7.63	0.0143
745.6	35.6	0.0154	**745.6**	19.5	0.0155	**745.6**	8.38	0.0157
838.8	38.7	0.0167	**838.8**	21.3	0.0169	**838.8**	9.12	0.0171
932	41.8	0.0181	**932**	23.0	0.0183	**932**	9.87	0.0185
1025.2	45	0.0194	**1025.2**	24.8	0.0197	**1025.2**	10.6	0.0199
1118.4	48.1	0.0208	**1118.4**	26.6	0.0211	**1118.4**	11.4	0.0213
1211.6	52.1	0.0221	**1211.6**	28.7	0.0229	**1211.6**	12.1	0.0227
Discount rate (%)								
9.775	41.2	0.0141	**9.775**	23.2	0.0147	**9.775**	9.84	0.0143
10.350	41.4	0.0154	**10.350**	23.4	0.0161	**10.350**	9.66	0.0155
10.925	41.7	0.0168	**10.925**	23.6	0.0176	**10.925**	9.78	0.0170
11.5	41.8	0.0181	**11.5**	23.0	0.0183	**11.5**	9.87	0.0185
12.075	41.8	0.0196	**12.075**	23.2	0.0199	**12.075**	9.94	0.0201
12.650	41.9	0.0209	**12.650**	23.4	0.0216	**12.650**	9.99	0.0217
13.225	42.1	0.0226	**13.225**	23.5	0.0223	**13.225**	10.0	0.0235
Inflation rate (%)								
13.345	42.8	0.0249	**13.345**	23.7	0.0255	**13.345**	10.0	0.0253
14.130	42.6	0.0225	**14.130**	23.6	0.0230	**14.130**	10.0	0.0229
14.915	42.3	0.0202	**14.915**	23.5	0.0207	**14.915**	9.95	0.0206
15.7	41.8	0.0181	**15.7**	23.0	0.0183	**15.7**	9.87	0.0185
18.84	40.9	0.0119	**18.84**	23.0	0.0123	**18.84**	10.6	0.0136
22.608	39.9	0.00703	**22.608**	26.0	0.00853	**22.608**	9.42	0.00714
27.130	39.0	0.00367	**27.130**	27.9	0.00487	**27.130**	12.8	0.00529
Scaled solar irradiance (kWh/m^2)								
3.5105	51.4	0.0231	**3.5105**	29.7	0.0251	**3.5105**	11.1	0.0206
3.7170	46.7	0.0205	**3.7170**	27.3	0.0226	**3.7170**	10.7	0.0198
3.9235	43.4	0.0189	**3.9235**	25.1	0.0203	**3.9235**	10.2	0.0190
4.13	41.8	0.0181	**4.13**	23.0	0.0183	**4.13**	9.87	0.0185
4.3365	40.3	0.0174	**4.3365**	22.9	0.0185	**4.3365**	9.69	0.0183
4.543	39.1	0.0169	**4.543**	22.6	0.0184	**4.543**	9.27	0.0172
4.7495	37.8	0.0164	**4.7495**	22.8	0.0187	**4.7495**	8.89	0.0166
Electrical demand (kWh/day)								
79,615	40.1	0.0185	**39,344**	22.8	0.0244	**16,666**	9.84	0.0229
75,425	38.9	0.0191	**41,658**	22.4	0.0203	**17,646**	9.42	0.0201
71,234	39.3	0.0209	**43,973**	22.7	0.0193	**18,627**	9.45	0.0188
83,805	41.8	0.0181	**46,287**	23.0	0.0183	**19,607**	9.87	0.0185
87,995	47.0	0.0200	**48,601**	26.0	0.0203	**20,587**	11.1	0.0205
92,186	52.4	0.0219	**50,916**	28.9	0.0221	**21,568**	12.4	0.0223
96,376	57.7	0.0237	**53,230**	31.9	0.0240	**22,548**	13.6	0.0242

evaluations, particularly in Nigeria's context characterized by relatively elevated and fluctuating inflation rates stemming from the removal of oil subsidies and the devaluation of the naira. Within this sensitivity analysis framework, adjustments were made to PV costs by ±10 % to accommodate price fluctuations within the Nigerian market. Concurrently, alterations is made to the inflation rate involving a reduction of 5 % and an augmentation of 20 %. Likewise, the discount rate was subject to adjustments of ±5 % to address significant fluctuations within the economic landscape. Moreover, the analysis concentrated on the scaled annual average solar irradiance and electrical demand as a fundamental technical variable, subjecting it to adjustments of ±5 % to accommodate variations in climate conditions and burgeoning population. Table 16 presents the results of the sensitivity analysis.

Sensitivity analysis of economic parameters

PV cost. The findings indicate that as the PV cost rises, both the NPC and LCOE increase, whereas a decrease in PV cost results in decreases in both NPC and LCOE. For Aluu, Table 16 illustrates that a 10 %, 20 % and 30 % increase from the current value of the PV cost results in a corresponding rise in NPC by 7.4 %, 13.1 % and 19.8 %, and in COE by 7.7 %, 13 %, and 18.1 %. Conversely, due to the economic competitiveness of PV modules, along with technological advancements and economies of scale, the cost of PV modules can decrease. In this scenario, a 10 % reduction in capital cost of PV modules diminishes the NPC and LCOE of the system by 7.4 % and 7.7 %, respectively. In Choba, when the inflation rate is at its lowest level, the NPC and COE are calculated, at $17.7 million and $0.0141/kWh, respectively. Conversely, the highest values of NPC and LCOE, $28.7 million and $0.0229/kWh, occur at the maximum inflation rate. A 10 %, 20 %, and 30 % increase in PV costs grew the NPC by 7.3 %, 15.7 %, and 19.9 %, as well as LCOE by 7.1 %, 15.3 %, and 20.1 %, whilst a 10 % decrease causes NPC and LCOE drops by 7.4 % and 7.7 %, respectively. For Rumuekini, with a PV cost for the optimized system set at $1025.2, the NPC and LCOE are $10.6 million and $0.0199/kWh, respectively. A 10 %, 20 %, and 30 % increase in PV costs leads to NPC and LCOE increments of 6.9 %, 13.4 % and 18.5 %, respectively. Beneficially, a 10 % reduction in PV costs leads to LCOE and NPC decreases of 7.6 % each. These variations are due to the system's heavy reliance on PV modules for electricity generation, directly linking the NPC and LCOE of the system to the cost of PV modules. Therefore, the system is notably sensitive to changes in PV module costs.

Discount rate. In Aluu, an increase in the discount rate leads to a rise in LCOE and slight increases in NPC, whereas a decrease in the discount rate results in a lower LCOE and slight decreases in NPC (Fig. 15a). In details, a reduction in the discount rate by 5 %, 10 %, and 15 % leads to a marginal decrease in NPC by 0.24 %, 0.96 %, and 1.4 %, respectively, and a reduction in COE by 7.2 %, 14.9 %, and 22.1 %. On the other hand, if the discount rate increases by 5 %, 10 %, and 15 %, the NPC experiences a slight increase of 0 %, 0.23 %, and 0.71 %, respectively, while the COE increases by 7.7 %, 13.4 %, and 19.9 %. For Choba, as illustrated in Fig. 15b, it is observed that discount rate below 10.35 % results in a reduction in LCOE and a lower NPC. The NPC experiences a slight increase when the discount rate ranges between 11.5 % and 13.225 %. The highest NPC values are observed at a discount rate of 10.925 %. In Rumuekini, an increase in the discount rate results in a rise in the LCOE, accompanied by only marginal increases in NPC. Conversely, a reduction in the discount rate leads to decreases in both LCOE and NPC, with the lowest NPC achieved at a discount rate of 10.35 % (Fig. 15c). The relative stability of the NPC indicates that the project's costs are structured to be less dependent on future cash flows, whereas the elevated LCOE reflects the heavier discounting of future benefits or cost offsets. Comparatively, in a study by Peirow et al. [53] a 4 % increase in the discount rate resulted in a 16 % increase in LCOE, whereas in this study, a 5 % increase raised the COE by 7.6 %, 8.0 %, and 8.0 % for Aluu, Choba, and Rumuekini, respectively.

Inflation rate. Table 16 illustrates that a higher inflation rate results in a decrease in the LCOE while only slightly increasing the NPC for both Aluu and Choba, and vice versa. Specifically, raising the inflation rate from 15.7 % to 27.13 % for Aluu notably reduces the LCOE, while the NPC shows a marginal change from $41.8 million to $39.0 million. Notably, if the inflation rate exceeds 22.608 %, LCOE stabilizes. In Rumuekini, increasing the inflation rate from 15.7 % to 27.130 % lowers COE from $0.0185/kWh to $0.00529/kWh, whereas decreasing the inflation rate from 15.7 % to 13.345 % raises LCOE to $0.0253/kWh, with minimal NPC variations. Moreso, adjusting inflation rates to 18.84 % and 27.13 % also impacts the size of PV modules used in optimized systems. This demonstrates that projects can become more viable in high-inflation economies. Inflation typically raises nominal costs of materials, labour, and other inputs, potentially increasing project costs initially. However, considering the time value of money, inflation can sometimes enhance the feasibility of long-term projects. Higher inflation rates reduce the discounted payback period, accelerating the financial viability of projects by discounting future cash flows less. This benefits investors with substantial debts, as they repay loans with devalued money. Despite complicating short-term economic planning and raising costs, inflation can offer strategic advantages for long-term investments.

Sensitivity analysis of technical parameters

Scaled solar irradiance. The effect of scaled solar irradiance on NPC and LCOE varies significantly depending on the intensity of solar radiation. Generally, an increase in scaled solar irradiance tends to decrease both NPC and COE, while a decrease in solar irradiance results in an increase in NPC and LCOE. For Aluu, higher solar radiation levels lead to adjustments in PV module capacity within optimized systems. More precisely, when the solar irradiation is increased by 5 %, 10 %, and 15 %, the NPC and LCOE generally reduce by 3.6 % and 3.9 %, 6.5 % and 6.6 %, and 9.6 % and 9.4 %, respectively. In Choba, a 5 %, 10 %, and 15 % decrease in solar radiation leads to increases in NPC and COE of 8.4 % and 10.9 %, 15.8 % and 19.0 %, and 22.6 % and 27.1 %. In Rumuekini, a 5 %, 10 %, and 15

% rise in solar radiation decreases the NPC and LCOE by 1.8 % and 1.1 %, 6.1 % and 7.0 %, and 9.9 % and 10.3 %, respectively. Conversely, a 5 %, 10 %, and 15 % decrease in solar radiation increases NPC and LCOE by 3.2 % and 2.7 %, 7.8 % and 6.6 %, and 11.1 % and 10.2 %, respectively. These adjustments made for Rumuekini impact the capacity of PV modules within the optimized systems. The effect of scaled solar irradiance on NPC and COE is crucial in the context of renewable energy systems, as it directly influences the economic feasibility and operational costs of solar power generation. Adjustments in solar irradiance levels affect the energy output of photovoltaic systems, thereby altering the NPC and LCOE.

Electrical demand. For the three communities understudy, variations in electrical demand directly impact the LCOE and NPC: an increase in electrical demand raises both LCOE and NPC, while a decrease in electrical demand increase LCOE without a clearly defined change in NPC. Increased load demand necessitates higher energy production, making it essential to update the capacity of different components in the hybrid system to boost overall electricity generation. Consequently, this leads to an increase in NPC and LCOE. Conversely, when electrical demand decreases, the system operates below its capacity, resulting in inefficiencies where infrastructure and investments are not fully utilized, leading to higher costs per unit of output. However, The NPC remains stable as it reflects the total investment and ongoing costs, not the efficiency of usage. According to the studies by [51], the NPC and LCOE rose by 10.7 % and 0.9 %, respectively, with a mere 0.09 % rise in load demand. Comparatively, this study for Aluu, Choba, and Rumuekini demonstrated that a 5 % rise in load demand led to an NPC and LCOE increase of 11.1 % and 0.19 %, 11.5 % and 0.2 %, and 11.1 % and 0.2 %, respectively. These findings underscore the project's viability.

Conclusion

Nigeria faces the formidable challenge of simultaneously ensuring a consistent provision of electricity, curbing greenhouse gas emissions, and preserving energy accessibility for consumers. This multifaceted issue is particularly acute in rural communities, which are disproportionately affected by the country's chronic energy shortages. These communities often experience unreliable or completely absent electricity supply, which hampers socioeconomic development and exacerbates poverty. Addressing these energy challenges requires a comprehensive strategy, with renewable energy emerging as a pivotal solution. This research study begins by reviewing the energy status to evaluate the energy situation in the rural area. It then investigates the techno-economic feasibility of a hybrid renewable energy system using HOMER Pro software. The study focuses on three proposed rural areas in Rivers State—Aluu, Choba, and Rumuekini—and is grounded in the utilization of renewable energy sources. Renewable energy resources (solar and wind), and battery were considered for design and simulation. Five different cases studied using hybrid energy systems in Aluu, Choba and Rumuekini were simulated and analysed by HOMER pro software. These results were further analysed to determine the best

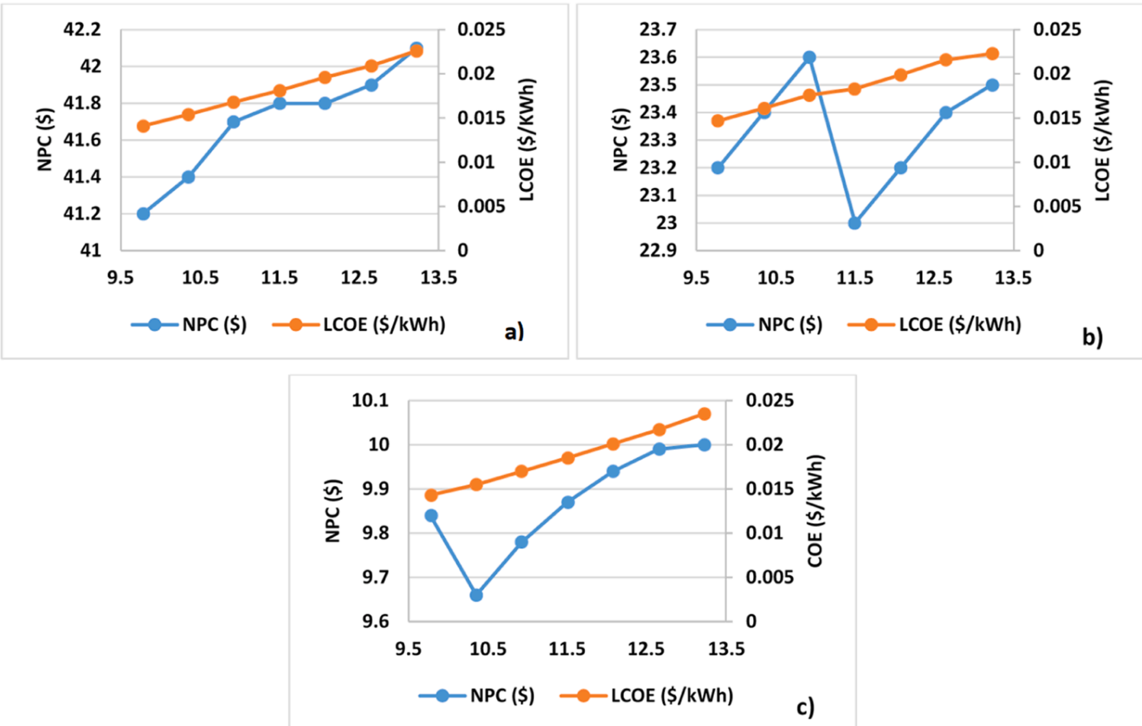

Fig. 15. Sensitivity analysis for variations in discount rate for a) Aluu b) Choba and c) Rumuekini.

combination for the location using the CRITIC-PROMETHEE II approach. The obtained results refer to the following outcomes:

- According to the results, residents of the Aluu, Choba, and Rumuekini communities, on average, receive 4.6, 5.2, and 6.1 h/day of electricity from the national grid. In terms of energy consumption in 2022, Aluu, Choba, and Rumuekini utilized 5436.4 MW, 3375.4 MW, and 1626.1 MW, respectively.
- Aluu and Rumuekini received the highest weightage of 0.202 and 0.186, respectively, for ease of installation (C_{s-2}) and for Choba, CO_2 emissions ($C_{\in-2}$) received the highest weightage of 0.208.
- The results of the MCDM analysis suggest that the grid/PV architecture (C_1) offers an appropriate trade-off solution for providing electricity to the Aluu, Choba, and Rumuekini communities.
- The optimal strategy for providing electricity to the Aluu community, considering the unreliable grid supply, entails the deployment of 33,500 kW PV modules and an 18,500 kW converter. This configuration results in an NPC of $41.8 million, a LCOE of $0.0181/kWh, and an operating cost of $10,295.
- For the Choba community, the optimization results recommend integrating 19,000 kW PV modules and a 9500 kW converter with the national grid. This optimized configuration results in an NPC of $23 million, a LCOE of $0.0183/kWh, and an operating cost of $2491.
- The optimal solution for electrifying the Rumuekini community involves the installation of 8000 kW PV modules and a 4000 kW converter. This optimized system configuration results in an NPC of $9.87 million, a LCOE of $0.0185/kWh, and an operating cost of $5044.
- The sensitivity analysis reveals that a rise in PV costs correlates with an increase in both the total NPC and COE, while a decrease in PV costs results in lower NPC and COE. For example, a 30 % increase in PV costs results in NPC and COE increases of 19.8 % and 18.1 % for Aluu, 19.9 % and 20.1 % for Choba and 18.4 % and 18.5 % for Rumuekini. Therefore, the system is notably sensitive to changes in PV module costs. Furthermore, the analysis shows that higher discount rates generally elevate the COE while causing slight increases in the NPC. Similarly, an increase in the inflation rate leads to higher total NPC and LCOE, again excluding the NPC for Choba and Rumuekini, which exhibit no clear pattern of change. The effect of scaled solar irradiance on NPC and LCOE varies significantly depending on the intensity of solar radiation. Generally, an increase in scaled solar irradiance tends to decrease both NPC and COE, while a decrease in solar irradiance results in an increase in NPC and LCOE. For the three communities understudy, variations in electrical demand directly impact the LCOE and NPC: an increase in electrical demand raises both LCOE and NPC, while a decrease in electrical demand increase LCOE without a clearly defined change in NPC.

This study provides essential tools for advancing rural electrification efforts, empowering stakeholders to design and implement sustainable, cost-effective, and socially inclusive energy solutions that improve the quality of life and promote economic development in rural communities. Furthermore, through this study, Nigeria can tackle its energy challenges while concurrently progressing towards its sustainable development objectives and contributing to global initiatives aimed at combating climate change. This result is also in line with the Nigerian Electricity Act which aims to liberalize the electricity sector, promote renewable energy sources, and decentralize power generation, transmission, and distribution.

Authorship statement

Conception and design of study: Chidozie Ezekwem, Suresh Muthusamy
Research and investigation: Chidozie Ezekwem, Suresh Muthusamy, Peace Chiamaka Ezekwem
Data curation, analysis and interpretation of data: Chidozie Ezekwem, Suresh Muthusamy, Peace Chiamaka Ezekwem
Drafting the manuscript: Chidozie Ezekwem
Revising the manuscript critically for important intellectual content: Chidozie Ezekwem

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] A. Acakpovi, P. Adjei, N. Nwulu, N.Y. Asabere, Optimal hybrid renewable energy system: a comparative study of wind/hydrogen/fuel-cell and wind/battery storage, J. Electr. Comput. Eng. 2020 (2020) 1–15.
[2] M.S. Adaramola, Viability of grid-connected solar PV energy system in Jos, Nigeria, Int. J. Electr. Power Energy Syst. 61 (2014) 64–69, https://doi.org/10.1016/j.ijepes.2014.03.015.
[3] M.S. Adaramola, S.S. Paul, O.M. Oyewola, Assessment of decentralized hybrid PV solar-diesel power system for applications in Northern part of Nigeria, Energy Sustainable Dev. 19 (2014) 72–82, https://doi.org/10.1016/j.esd.2013.12.007.
[4] B. Adebanji, G.A. Adepoju, P. Olulope, T. Fasina, O. Adetan, Feasibility and optimal design of a hybrid power system for rural electrification for a small village, Int. J. Electr. Comput. Eng. (IJECE) 10 (2020) 6214–6224.
[5] A. Alanazi, M. Alanazi, Multicriteria decision-making for evaluating solar energy source of Saudi Arabia, Sustainability 15 (2023) 10228.
[6] T. Ali, H. Ma, A.J. Nahian, A multi-criteria decision-making approach to determine the optimal hybrid energy system in coastal off-grid areas: a case study of Bangladesh, Process Integr. Optimiz. Sustainability 4 (2020) 265–277, https://doi.org/10.1007/s41660-020-00116-9.
[7] A.O. Amole, S. Oladipo, O.E. Olabode, K.A. Makinde, P. Gbadega, Analysis of grid/solar photovoltaic power generation for improved village energy supply: a case of Ikose in Oyo State Nigeria, Renewable Energy Focus 44 (2023) 186–211.

[8] B. Ariyo, M. Akorede, I. Omeiza, S. Amuda, S. Oladeji, Optimisation analysis of a stand-alone hybrid energy system for the senate building, university of Ilorin, Nigeria, J. Build. Eng. 19 (2018) 285–294.
[9] O.C. Aworh, Food safety issues in fresh produce supply chain with particular reference to sub-Saharan Africa, Food Control 123 (2021) 107737.
[10] O. Ayobami, G. Mark, Z. Kadri-Alabi, C.R. Achi, J.C. Jacob, COVID-19: an opportunity to re-evaluate the implementation of a One Health approach to tackling emerging infections in Nigeria and other sub-Saharan African countries, J. Egypt. Public Health Assoc. 96 (2021) 1–6.
[11] A.S. Aziz, M.F.N. Tajuddin, M.R. Adzman, M.F. Mohammed, M.A. Ramli, Feasibility analysis of grid-connected and islanded operation of a solar PV microgrid system: a case study of Iraq, Energy 191 (2020) 116591.
[12] O.M. Babatunde, D.E. Babatunde, I.H. Denwigwe, T.B. Adedoja, O.S. Adedoja, T.E. Okharedia, Analysis of an optimal hybrid power system for an off-grid community in Nigeria, Int. J. Energy Sector Manage. 14 (2020) 335–357.
[13] O.M. Babatunde, J.L. Munda, Y. Hamam, Hybridized off-grid fuel cell/wind/solar PV /battery for energy generation in a small household: a multi-criteria perspective, Int. J. Hydrogen. Energy 47 (2022) 6437–6452, https://doi.org/10.1016/j.ijhydene.2021.12.018.
[14] S. Bahramara, M.P. Moghaddam, M.R. Haghifam, Optimal planning of hybrid renewable energy systems using HOMER: a review, Renewable Sustainable Energy Rev. 62 (2016) 609–620.
[15] N. Bashir, B. Modu, P. Harcourt, Techo-economic analysis of off-grid renewable energy systems for rural electrification in North-eastern Nigeria, Int. J. Renewable Energy Res. (IJRER) 8 (2018) 1217–1228.
[16] J.-P. Brans, P. Vincke, Note—a preference ranking organisation method: (the PROMETHEE method for multiple criteria decision-making), Manage Sci. 31 (1985) 647–656.
[17] CBN, Exchange Rates [WWW Document], Central Bank of Nigeria, 2022. URL, https://www.cbn.gov.ng/rates/ExchRateByCurrency.asp.
[18] S.W. Chisale, S. Eliya, J. Taulo, Optimization and design of hybrid power system using HOMER pro and integrated CRITIC-PROMETHEE II approaches, Green Technol. Sustainability 1 (2023) 100005.
[19] O. Collins, Businesses Lose $29 Billion Yearly to Poor electricity, Says World Bank, TheGuardian, 2021.
[20] R.A. da Cunha, L.A.D. Rangel, C.A. Rudolf, L. dos Santos, A decision support approach employing the PROMETHEE method and risk factors for critical supply assessment in large-scale projects, Oper. Res. Perspect. 9 (2022) 100238, https://doi.org/10.1016/j.orp.2022.100238.
[21] K. Dassa, A. Recioui, Application of optimization to sizing renewable energy systems and energy management in microgrids: state of the art and trends, Appl. Nat.-Inspired Comput. Renewable Energy Syst. (2022) 60–94.
[22] F.Z. Dekhandji, M. Bouaraki, N. Zaidi, A. Recioui, Demand response and optimal sizing of hybrid renewable energy systems, Int. J. Control, Energy Electr. Eng. (ICEEE) 7 (2019) 35–40.
[23] D. Diakoulaki, G. Mavrotas, L. Papayannakis, Determining objective weights in multiple criteria problems: the critic method, Comput. Oper. Res. 22 (1995) 763–770, https://doi.org/10.1016/0305-0548(94)00059-H.
[24] E.O. Diemuodeke, A. Addo, C.O.C. Oko, Y. Mulugetta, M.M. Ojapah, Optimal mapping of hybrid renewable energy systems for locations using multi-criteria decision-making algorithm, Renew. Energy 134 (2019) 461–477, https://doi.org/10.1016/j.renene.2018.11.055.
[25] H. El-Houari, A. Allouhi, T. Salameh, T. Kousksou, A. Jamil, B. El Amrani, Energy, Economic, Environment (3E) analysis of WT-PV-Battery autonomous hybrid power plants in climatically varying regions, Sustainable Energy Technol. Assess. 43 (2021) 100961.
[26] Esan, A.B., Agbetuyi, A.F., Oghorada, O., Ogbeide, K., Awelewa, Ayokunle.A., Afolabi, A.E., 2019. Reliability assessments of an islanded hybrid PV-diesel-battery system for a typical rural community in Nigeria. Heliyon. 5, e01632. https://doi.org/10.1016/j.heliyon.2019.e01632.
[27] C. Ezekwem, S. Muthusamy, Feasibility study of integrating the renewable energy system for increased electricity access: a case study of Choba community in Nigeria, Sci. Afr. (2023) e01781.
[28] A. Giwa, A. Alabi, A. Yusuf, T. Olukan, A comprehensive review on biomass and solar energy for sustainable energy generation in Nigeria, Renewable Sustainable Energy Rev. 69 (2017) 620–641, https://doi.org/10.1016/j.rser.2016.11.160.
[29] M. He, H. Forootan Fard, K. Yahya, M. Mohamed, I. Alhamrouni, L.J. Awalin, Optimal design of hybrid renewable systems, including grid, PV, bio generator, diesel generator, and battery, Sustainability 15 (2023) 3297.
[30] M.H. Jahangir, R. Cheraghi, Economic and environmental assessment of solar-wind-biomass hybrid renewable energy system supplying rural settlement load, Sustainable Energy Technol. Assess. 42 (2020) 100895, https://doi.org/10.1016/j.seta.2020.100895.
[31] M. Jahangiri, A.A. Shamsabadi, A. Mostafaeipour, M. Rezaei, Y. Yousefi, L.M. Pomares, Using fuzzy MCDM technique to find the best location in Qatar for exploiting wind and solar energy to generate hydrogen and electricity, Int. J. Hydrogen. Energy 45 (2020) 13862–13875, https://doi.org/10.1016/j.ijhydene.2020.03.101.
[32] S. Jain, Y. Sawle, Optimization and comparative economic analysis of standalone and grid-connected hybrid renewable energy system for remote location, Front. Energy Res. 9 (2021) 724162.
[33] C.A. John, L.S. Tan, J. Tan, P.L. Kiew, A.M. Shariff, H.N. Abdul Halim, Selection of renewable energy in rural area via life cycle assessment-analytical hierarchy process (LCA-AHP): a case study of Tatau, Sarawak, Sustainability 13 (2021) 11880.
[34] I.A. Jumare, R. Bhandari, A. Zerga, Assessment of a decentralized grid-connected photovoltaic (PV)/wind/biogas hybrid power system in northern Nigeria, Energy Sustain. Soc. 10 (2020) 1–25.
[35] J. Jurasz, M. Guezgouz, P.E. Campana, A. Kies, On the impact of load profile data on the optimization results of off-grid energy systems, Renewable Sustainable Energy Rev. 159 (2022) 112199, https://doi.org/10.1016/j.rser.2022.112199.
[36] Y.W. Koholé, C.A.W. Ngouleu, F.C.V. Fohagui, G. Tchuen, Optimization of an off-grid hybrid photovoltaic/wind/diesel/fuel cell system for residential applications power generation employing evolutionary algorithms, Renew. Energy (2024) 120131.
[37] Y.W. Koholé, C.A. Wankouo Ngouleu, F.C.V. Fohagui, G. Tchuen, Quantitative techno-economic comparison of a photovoltaic/wind hybrid power system with different energy storage technologies for electrification of three remote areas in Cameroon using Cuckoo search algorithm, J. Energy Storage 68 (2023) 107783, https://doi.org/10.1016/j.est.2023.107783.
[38] M. Madkour, Electricity in Egypt:"From crisis into phenomenon, in: Presented at the 2021 22nd International Middle East Power Systems Conference (MEPCON), IEEE, 2021, pp. xviii–xvixx.
[39] Mahajan, A., Binaz, V., Singh, I., Arora, N., 2022. Selection of natural fiber for sustainable composites using hybrid multi criteria decision making techniques. Compos. Part C: Open Access 7, 100224. https://doi.org/10.1016/j.jcomc.2021.100224.
[40] J. Markovic Brankovic, M. Markovic, D. Nikolic, Comparative study of hydraulic structures alternatives using promethee II complete ranking method, Water Resour. Manage. 32 (2018) 3457–3471, https://doi.org/10.1007/s11269-018-2001-x.
[41] A.A. Masud, The application of homer optimization software to investigate the prospects of hybrid renewable energy system in rural communities of Sokoto in Nigeria, Int. J. Electr. Comput. Eng. (IJECE) 7 (2017) 596–603.
[42] A.A. Mas'ud, A.V. Wirba, F. Muhammad-Sukki, I.A. Mas'ud, A.B. Munir, N.M. Yunus, An assessment of renewable energy readiness in Africa: case study of Nigeria and Cameroon, Renewable Sustainable Energy Rev. 51 (2015) 775–784.
[43] D. Misra, Current status and future potential of solar energy utilization in Rajasthan, India, in: Smart Energy and Advancement in Power Technologies: Select Proceedings of ICSEAPT 2021 1, Springer, 2022, pp. 697–710.
[44] G. Mohammed, A. Gado, U.A. Bello, Feasibility study of a stand-alone wind-solar hybrid energy system for off-grid rural electrification in Sokoto State, Nigeria, Sci. Technol. 6 (2020) 64–72.
[45] C.A. Nallolla, V. Perumal, Optimal design of a hybrid off-grid renewable energy system using techno-economic and sensitivity analysis for a rural remote location, Sustainability 14 (2022) 15393.
[46] O.A. Odetoye, P.K. Olulope, O.M. Olanrewaju, A.O. Alimi, O.G. Igbinosa, Multi-year techno-economic assessment of proposed zero-emission hybrid community microgrid in Nigeria using HOMER, Heliyon 9 (2023).
[47] K. Ogunyale, Nigeria's National Grid Provides Only 51.2% Of Electricity, TwentyTenDaily, 2021.

[48] O.D. Ohijeagbon, A. Oluseyi, O. Waheed, M. Adekojo, E.Y. Salawu, F.A. Oyawale, Design of optimal hybrid renewable energy system for sustainable power supply to isolated-grid communities in North Central, Nigeria, Procedia Manuf. 35 (2019) 278–284.
[49] A.S. Okoh, E. Okpanachi, Transcending energy transition complexities in building a carbon-neutral economy: the case of Nigeria, Clean. Energy Syst. 6 (2023) 100069.
[50] P.C. Okonkwo, E.M. Barhoumi, W. Emori, M.I. Shammas, P.C. Uzoma, A.M. Amer Mohamed, A.M. Abdullah, Economic evaluation of hybrid electrical systems for rural electrification: a case study of a rural community in Nigeria, Int. J. Green. Energy 19 (2022) 1059–1071.
[51] J.O. Oladigbolu, M.A. Ramli, Y.A. Al-Turki, Techno-economic and sensitivity analyses for an optimal hybrid power system which is adaptable and effective for rural electrification: a case study of Nigeria, Sustainability 11 (2019) 4959.
[52] L. Olatomiwa, S. Mekhilef, A.S.N. Huda, O.S. Ohunakin, Economic evaluation of hybrid energy systems for rural electrification in six geo-political zones of Nigeria, Renew. Energy 83 (2015) 435–446, https://doi.org/10.1016/j.renene.2015.04.057.
[53] S. Peirow, F. Razi Astaraei, A. Saifoddin Asl, Techno-economic and environmental assessment of a hybrid renewable energy system for a hospital using multi-criteria decision-making method, Energies 16 (2023) 1916.
[54] R. Rajbongshi, D. Borgohain, S. Mahapatra, Optimization of PV-biomass-diesel and grid base hybrid energy systems for rural electrification by using HOMER, Energy 126 (2017) 461–474.
[55] H. Rezk, B. Alamri, M. Aly, A. Fathy, A.G. Olabi, M.A. Abdelkareem, H.A. Ziedan, Multicriteria decision-making to determine the optimal energy management strategy of hybrid PV–diesel battery-based desalination system, Sustainability 13 (2021) 4202.
[56] T.M.I. Riayatsyah, T.A. Geumpana, I.M.R. Fattah, S. Rizal, T.M.I. Mahlia, Techno-economic analysis and optimisation of campus grid-connected hybrid renewable energy system using HOMER grid, Sustainability (2022) 14, https://doi.org/10.3390/su14137735.
[57] H.M. Ridha, H. Hizam, S. Mirjalili, M.L. Othman, M.E. Ya'acob, M. Ahmadipour, Innovative hybridization of the two-archive and PROMETHEE-II triple-objective and multi-criterion decision making for optimum configuration of the hybrid renewable energy system, Appl. Energy 341 (2023) 121117.
[58] T. Salameh, E.T. Sayed, M.A. Abdelkareem, A.G. Olabi, H. Rezk, Optimal selection and management of hybrid renewable energy system: Neom city as a case study, Energy Convers. Manage 244 (2021) 114434, https://doi.org/10.1016/j.enconman.2021.114434.
[59] S. Salisu, M.W. Mustafa, L. Olatomiwa, O.O. Mohammed, Assessment of technical and economic feasibility for a hybrid PV-wind-diesel-battery energy system in a remote community of north central Nigeria, Alexandria Eng. J. 58 (2019) 1103–1118, https://doi.org/10.1016/j.aej.2019.09.013.
[60] M.M.A. Seedahmed, M.A.M. Ramli, H.R.E.H. Bouchekara, A.H. Milyani, M. Rawa, F.N. Budiman, R.F. Muktiadji, S.M.U. Hassan, Optimal sizing of grid-connected photovoltaic system for a large commercial load in Saudi Arabia, Alexandria Eng. J. 61 (2022) 6523–6540, https://doi.org/10.1016/j.aej.2021.12.013.
[61] I. Sofimieari, M.W.B. Mustafa, F. Obite, Modelling and analysis of a PV/wind/diesel hybrid standalone microgrid for rural electrification in Nigeria, Bull. Electr. Eng. Inform. 8 (2019) 1468–1477.
[62] Statista, 2024. Nigeria: power production share by source 2023 [WWW Document]. URL https://www.statista.com/statistics/1237541/nigeria-distribution-of-electricity-production-by-source/#:~:text=Natural%20gas%20is%20by%20far,percent%20of%20Nigeria's%20power%20production.
[63] Statista, 2023. Renewable energy share of electricity capacity in Nigeria from 2011 to 2022 [WWW Document]. URL https://www.statista.com/statistics/1278245/renewable-energy-share-of-electricity-capacity-in-nigeria/#:~:text=As%20of%202022%2C%20renewable%20energy,decrease%20over%20the%20past%20years. (accessed 12.29.23).
[64] Statista, 2023. Lithium-ion battery price worldwide from 2013 to 2023(in 2023 U.S. dollars per kilowatt-hour) Price in 2023 U.S. dollars per kilowatt-hour.
[65] TheCable, Nigeria's National Grid Suffered 222 System Collapses in 12 Years, TheCable, 2022.
[66] H.O. Tijani, C. Wei Tan, N. Bashir, Techno-economic analysis of hybrid photovoltaic/diesel/battery off-grid system in northern Nigeria, J. Renewable Sustainable Energy 6 (2014).
[67] S. Twaha, M.H. Idris, M. Anwari, A. Khairuddin, Applying grid-connected photovoltaic system as alternative source of electricity to supplement hydro power instead of using diesel in Uganda, Energy 37 (2012) 185–194.
[68] M.O. Ukoba, O.E. Diemuodeke, M. Alghassab, H.I. Njoku, M. Imran, Z.A. Khan, Composite multi-criteria decision analysis for optimization of hybrid renewable energy systems for geopolitical zones in Nigeria, Sustainability 12 (2020) 5732.
[69] Z. Ullah, M.R. Elkadeem, K.M. Kotb, I.B.M. Taha, S. Wang, Multi-criteria decision-making model for optimal planning of on/off grid hybrid solar, wind, hydro, biomass clean electricity supply, Renew. Energy 179 (2021) 885–910, https://doi.org/10.1016/j.renene.2021.07.063.
[70] M. Ur Rashid, I. Ullah, M. Mehran, M.N.R. Baharom, F. Khan, Techno-economic analysis of grid-connected hybrid renewable energy system for remote areas electrification using homer pro, J. Electr. Eng. Technol. 17 (2022) 981–997.
[71] C.A. Wankouo Ngouleu, Y.W. Koholé, F.C.V. Fohagui, G. Tchuen, Techno-economic analysis and optimal sizing of a battery-based and hydrogen-based standalone photovoltaic/wind hybrid system for rural electrification in Cameroon based on meta-heuristic techniques, Energy Convers. Manage 280 (2023) 116794, https://doi.org/10.1016/j.enconman.2023.116794.
[72] World Bank, 2022. Access to Electricity (% of Population) - Nigeria [WWW Document]. Data.worldbank.org. URL https://data.worldbank.org/indicator/EG.ELC.ACCS.ZS?locations=NG (accessed 4.10.23).
[73] WorldBank, 2024. Access to Electricity, Rural (% of Rural Population) [WWW Document]. World Bank Open Data. URL https://data.worldbank.org/indicator/EG.ELC.ACCS.RU.ZS.
[74] K. Zorlu, V. Dede, Evaluation of nature-based tourism potential in protected and sensitive areas by CRITIC and PROMETHEE-GAIA methods, Int. J. Geoheritage Parks 11 (2023) 349–364, https://doi.org/10.1016/j.ijgeop.2023.05.004.